



CH516 – Advanced Process Control Evaporator System

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OBJECTIVE :- Maintain Desired concentration X_2

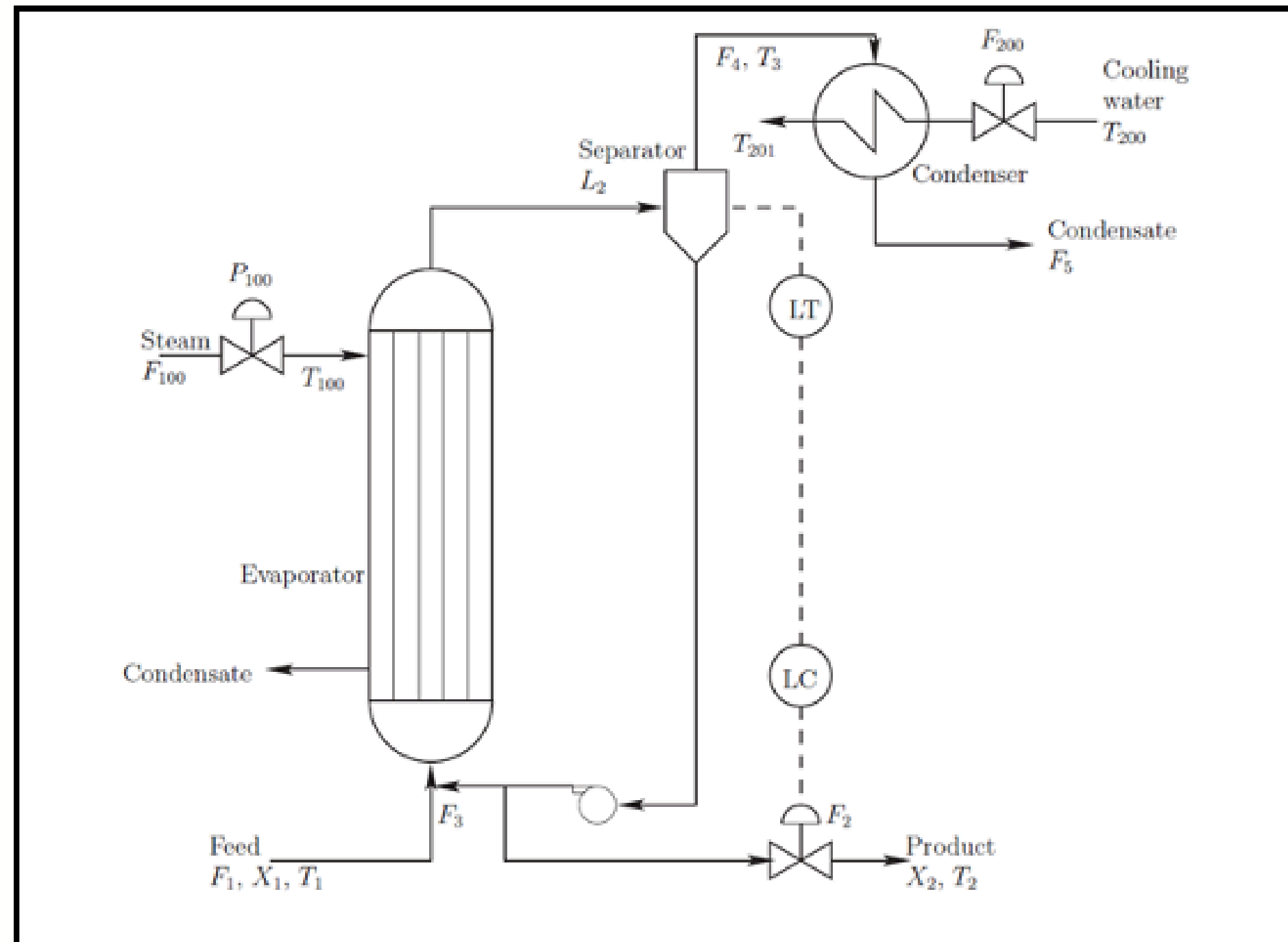


Fig - 1 :- Evaporator System

Introduction of the Allotted Process

1) State Variables

$$x = [X_2 \quad P_2]$$

Where:

- X_2 = Product concentration
- P_2 = Operating pressure

2) Manipulated Variables

$$u = [P_{100} \quad F_{200}]$$

Where:

- P_{100} = Steam pressure
- F_{200} = Cooling water flow rate

3) Disturbances

- F_1 = Feed flow rate
- X_1 = Feed concentration
- F_3 = Circulation flow rate
- T_1 = Feed temperature
- T_{200} = Cooling water inlet temperature

4) Mass Balance

$$M (dX_2/dt) = F_1X_1 - F_2X_2$$

5) Pressure Dynamics

$$C (dP_2/dt) = F_4 - F_5$$

Where:

- F_4 = Vapor generation rate
- F_5 = Condensation rate

Additional Equations

1) Process Liquid Energy Balance

$$T_2 = 0.5616P_2 + 0.3126X_2 + 48.43$$

$$T_3 = 0.507P_2 + 55$$

$$F_4 = (Q_{100} - F_1 C_p (T_2 - T_1)) / \lambda$$

2) Heat Steam Jacket

$$T_{100} = 0.1538P_{100} + 90$$

$$Q_{100} = UA_1(T_{100} - T_2)$$

$$UA_1 = 0.16(F_1 + F_3)$$

$$F_{100} = Q_{100} / \lambda_s$$

3) Condenser

$$Q_{200} = (UA_2(T_3 - T_{200})) / (1 + (2C_p F_{200} / UA_2))$$

$$F_5 = Q_{200} / \lambda$$

4) Level Controller

$$F_2 = F_1 - F_4$$

OBJECTIVE

MPC Objective Function :- Minimize this objective function

$$J = \sum_{k=0}^{N-1} [(\mathbf{x} - \mathbf{x}_s)^T \mathbf{Q} (\mathbf{x} - \mathbf{x}_s) + (\mathbf{u} - \mathbf{u}_s)^T \mathbf{R} (\mathbf{u} - \mathbf{u}_s) + \Delta \mathbf{u}^T \mathbf{S} \Delta \mathbf{u}]$$

Control Objectives

- **Maintain desired product concentration**
- Maintain operating pressure
- Reject disturbances
- Ensure smooth control action
- Satisfy process constraints

Process Constraints

$$\begin{aligned} X_2 &\geq 25 \\ 40 &\leq P_2 \leq 80 \\ P_{100} &\leq 400 \\ F_{200} &\leq 400 \end{aligned}$$

Linearized Model and Stability Analysis

1) Steady-State Operating Point

$$\mathbf{x}_{ss} = [25.0000 \quad 50.5700] = [X2_{ss} \quad P2_{ss}]$$

2) Continuous-Time State Space Model

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{D}\mathbf{d}$$

3) State Matrix (A)

$$\mathbf{A} = \begin{bmatrix} -0.2074 & -0.1878 \\ -0.0209 & -0.0558 \end{bmatrix}$$

4) Input Matrix (B)

$$\mathbf{B} = \begin{bmatrix} 0.0479 & 0 & -0.9191 & 0.5000 & 0.1824 & 0.0227 & 0 \\ 0.0096 & -0.0018 & 0.0162 & 0 & 0.0365 & 0.0045 & 0.0359 \end{bmatrix}$$

Linearized Model and Stability Analysis

Stability Analysis (Continuous System)

1) Eigenvalues of A:

$$\lambda(A) = [-0.2300 \quad -0.0332]$$

[System is asymptotically stable]

Discrete-Time Model

$$x(k+1) = \Phi x(k) + \Gamma_U u(k) + \Gamma_D d(k)$$

2) State Transition Matrix (Φ)

$$\Phi = [\begin{array}{cc} 0.8144 & -0.1649 \\ -0.0184 & 0.9475 \end{array}]$$

3) Input Matrix (Γ_U)

$$\Gamma_U = [\begin{array}{cc} 0.0425 & 0.0002 \\ 0.0089 & -0.0018 \end{array}]$$

3) Disturbance Matrix (Γ_D)

$$\Gamma_D = [\begin{array}{cccc} -0.8320 & 0.4518 & 0.1617 & 0.0201 \\ -0.0031 & 0.0246 & -0.0048 & 0.0338 \\ & 0.0350 & 0.0042 & \end{array}]$$

Stability Analysis (Discrete System)

1) Eigenvalues of Φ :

$$\lambda(\Phi) = [0.7946 \quad 0.9673]$$

Since all eigenvalues lie inside the unit circle:
[Discrete system is asymptotically stable]

MPC Formulation – Varying Horizons

The Model Predictive Controller (MPC) was implemented for both:

- Disturbance Rejection
- Setpoint Tracking

The controller performance was analyzed by varying the:

- Prediction Horizon (N_p)
- Control Horizon (N_c)

Horizons Considered

1) Prediction Horizons

$$N_p = [20, 30]$$

2) Control Horizons

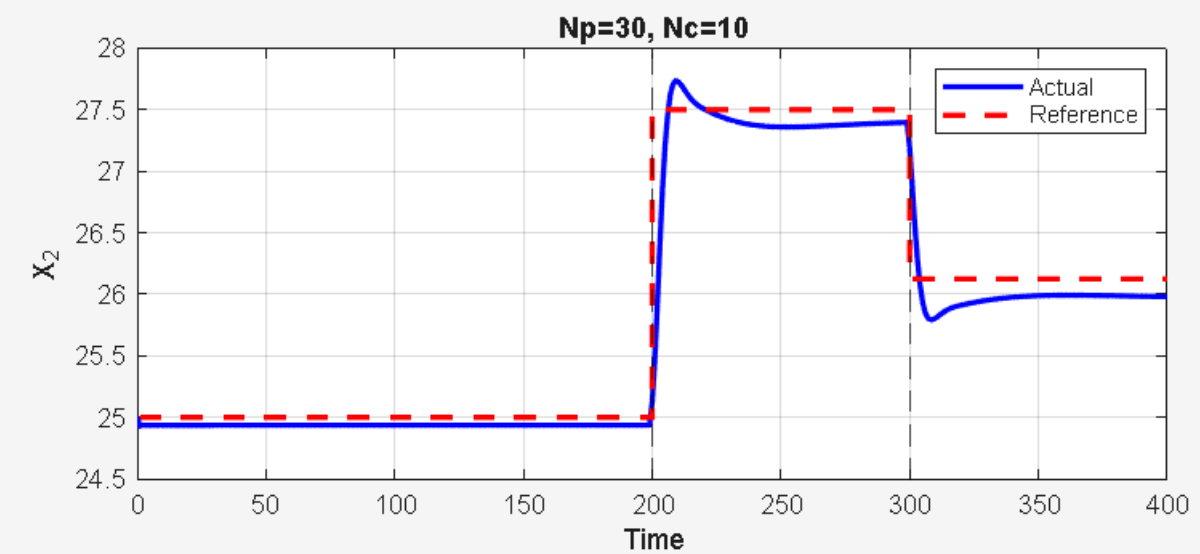
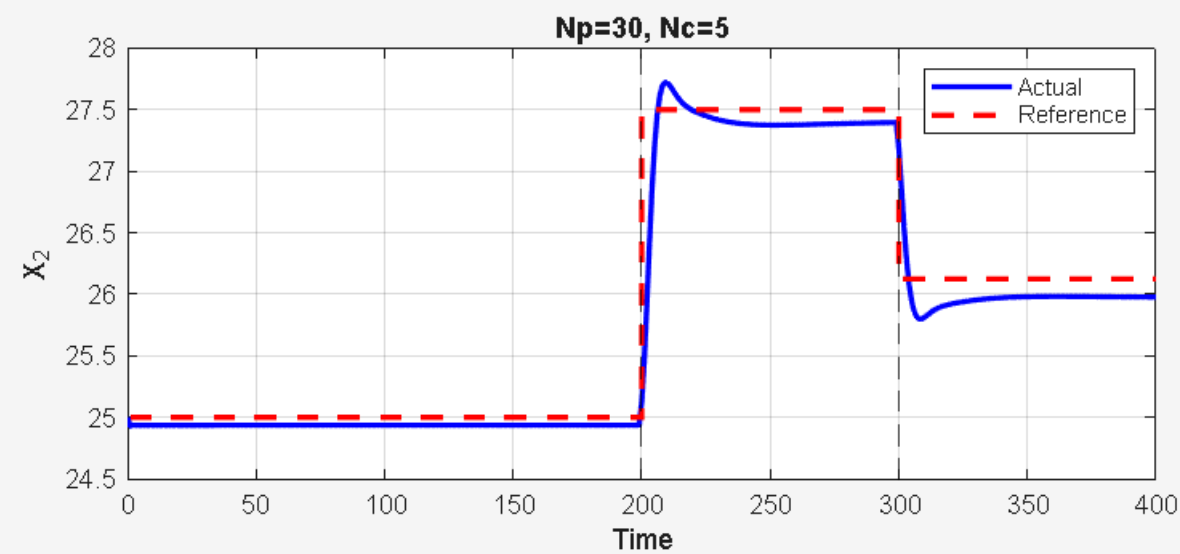
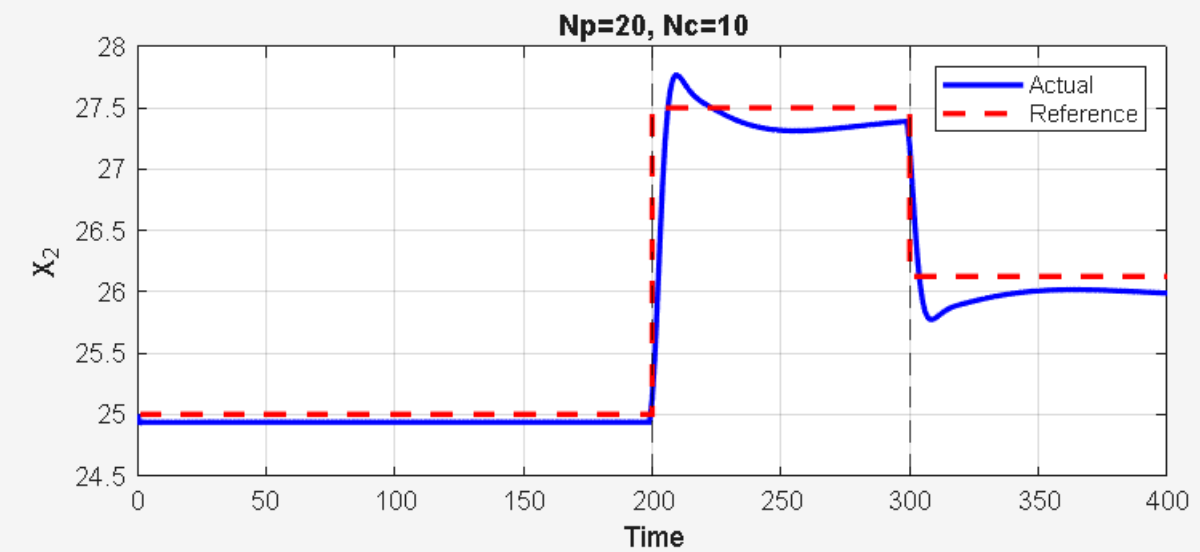
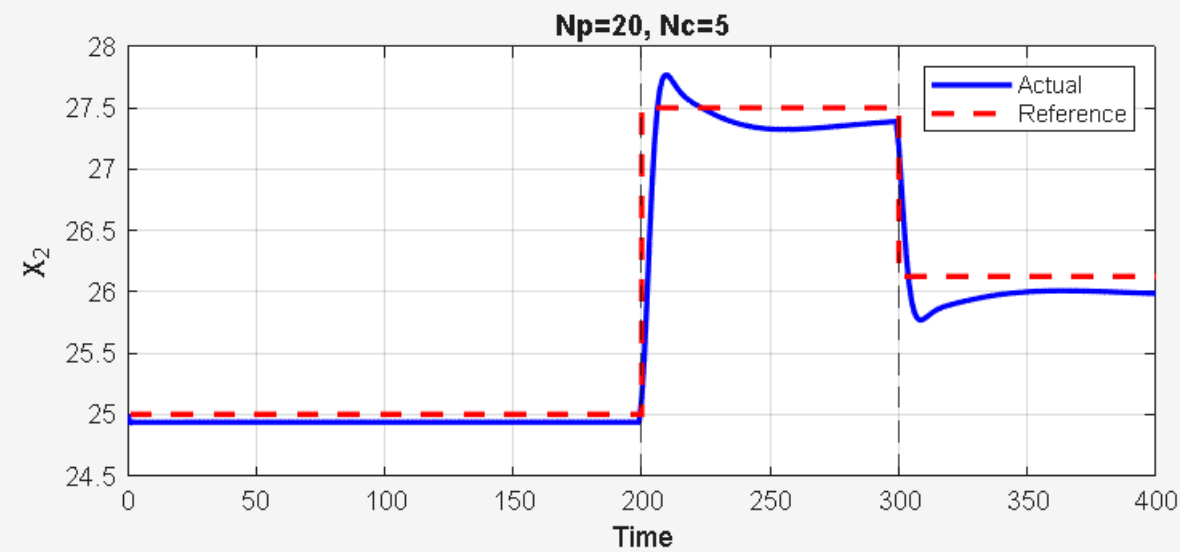
$$N_c = [5, 10]$$

$$Q = \begin{bmatrix} 20 & 0 \\ 0 & 3 \end{bmatrix}$$

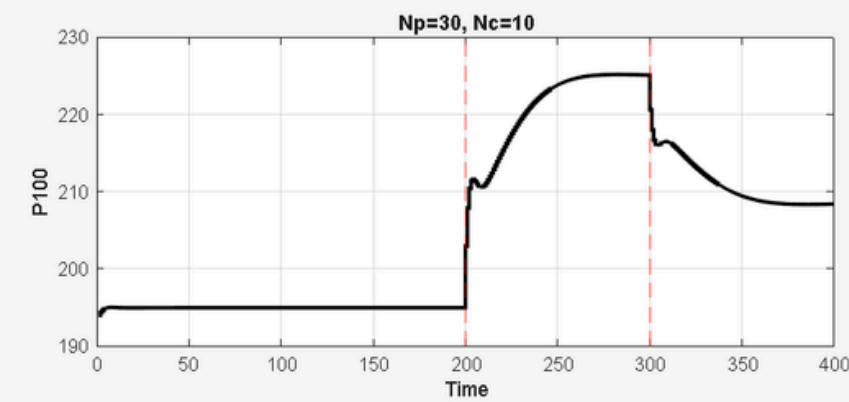
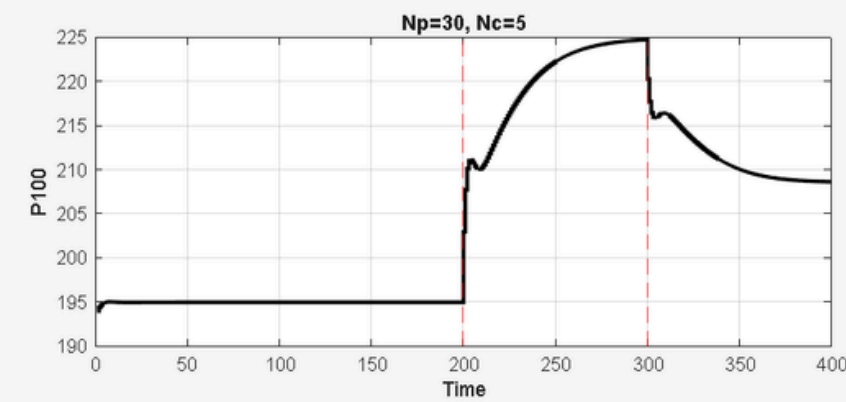
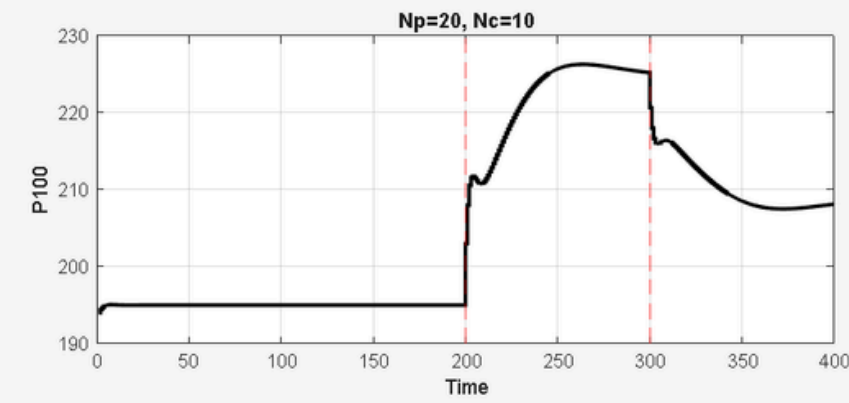
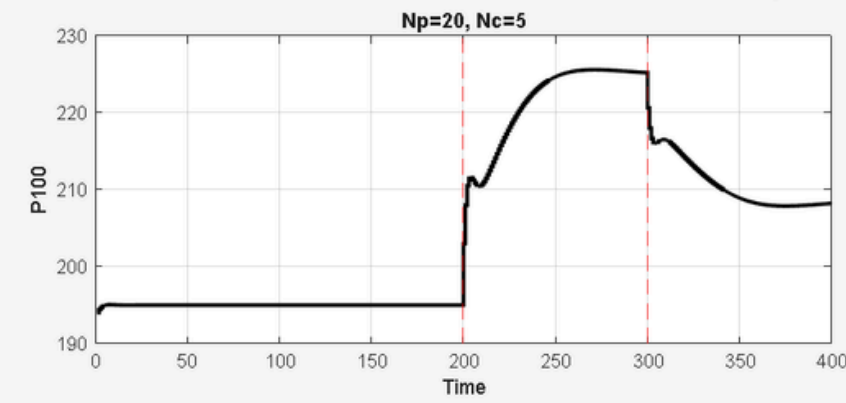
$$R = \begin{bmatrix} 1.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$

LMPC Set Point Tracking - Varying Horizons

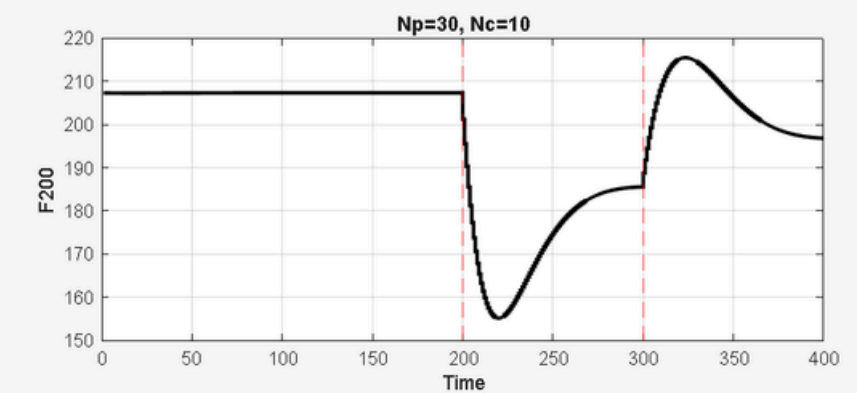
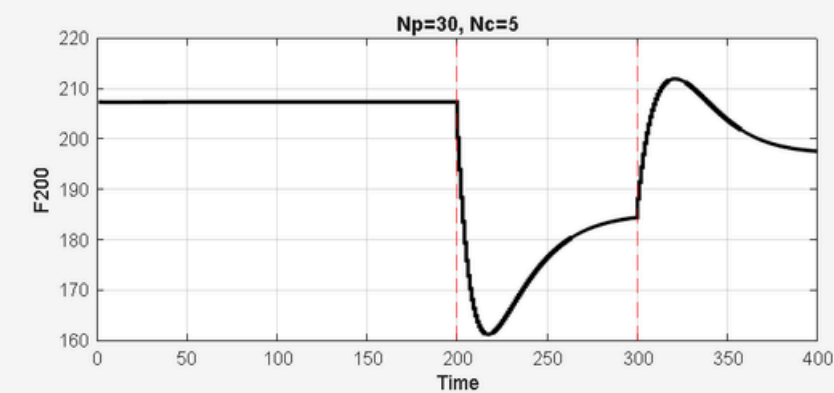
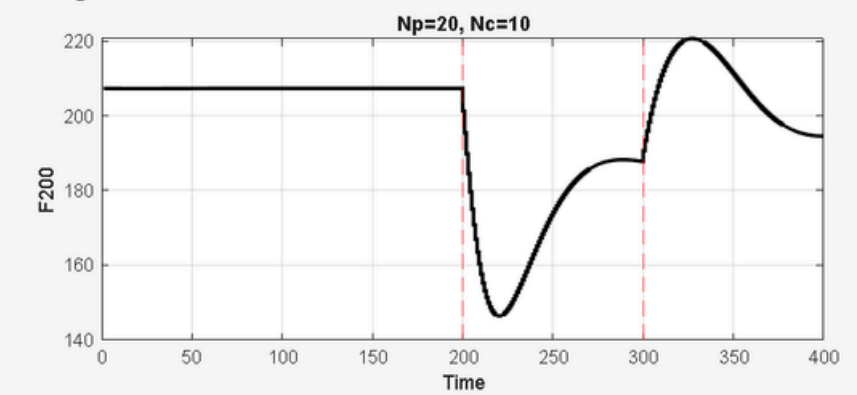
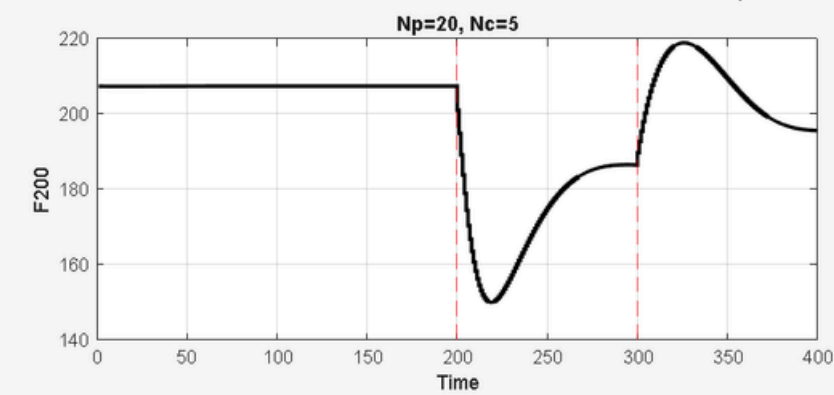
LMPC Setpoint Tracking : Product Composition



NMPC Setpoint Tracking : Steam Pressure

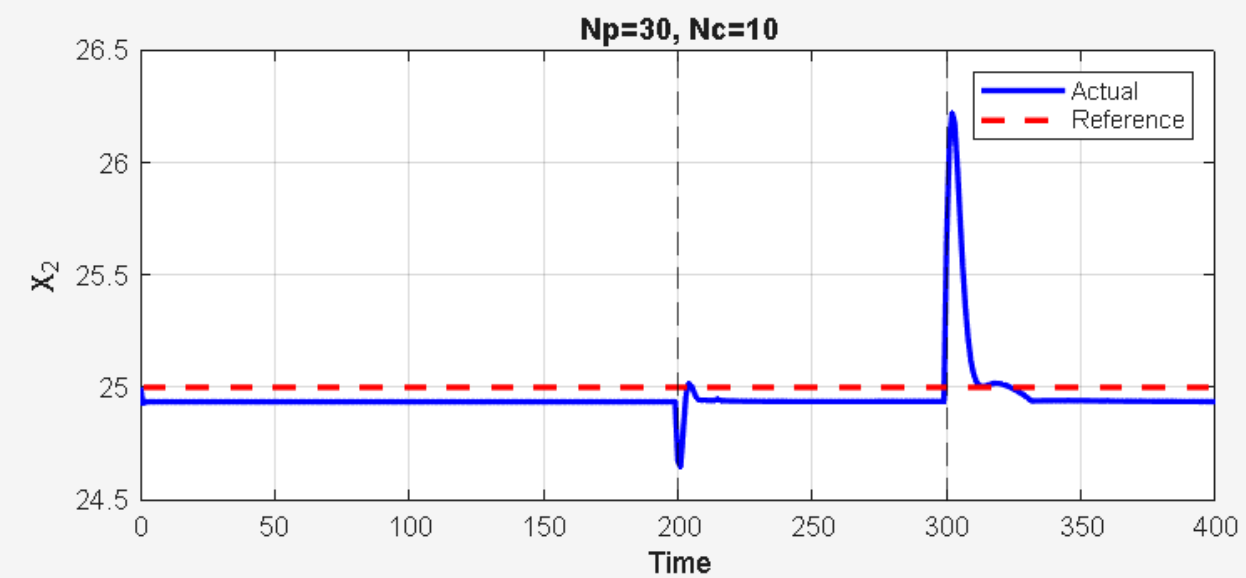
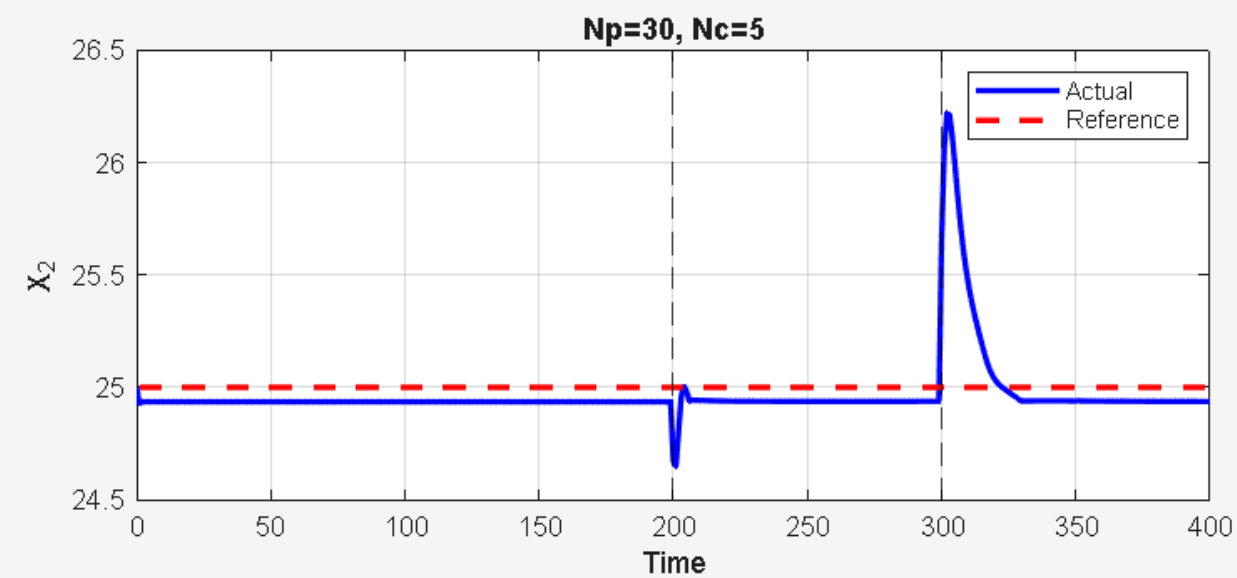
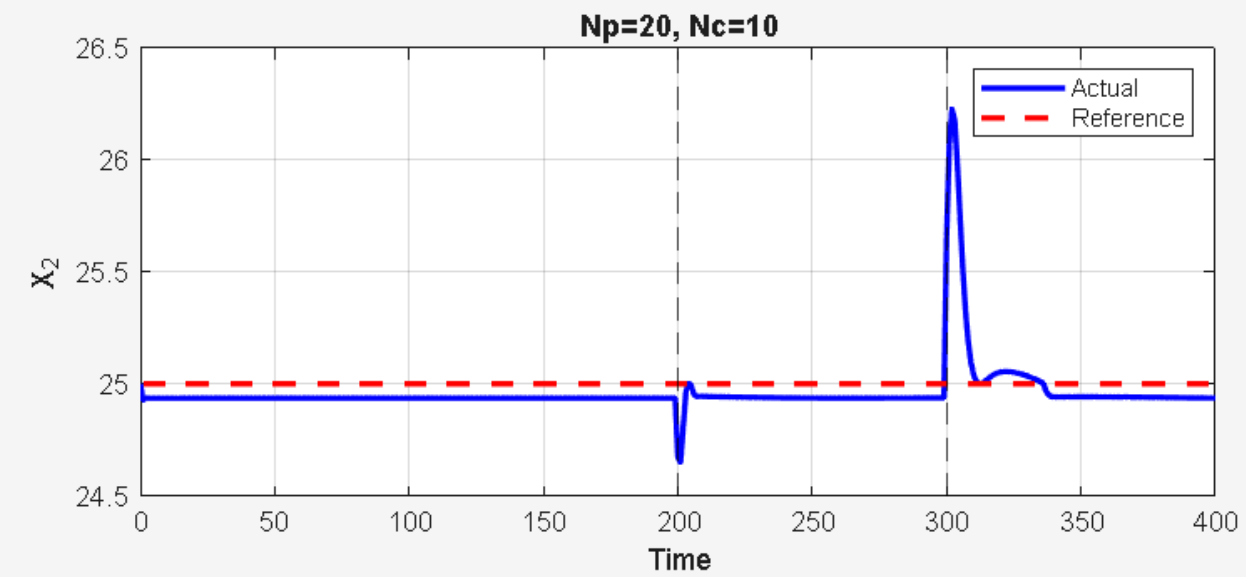
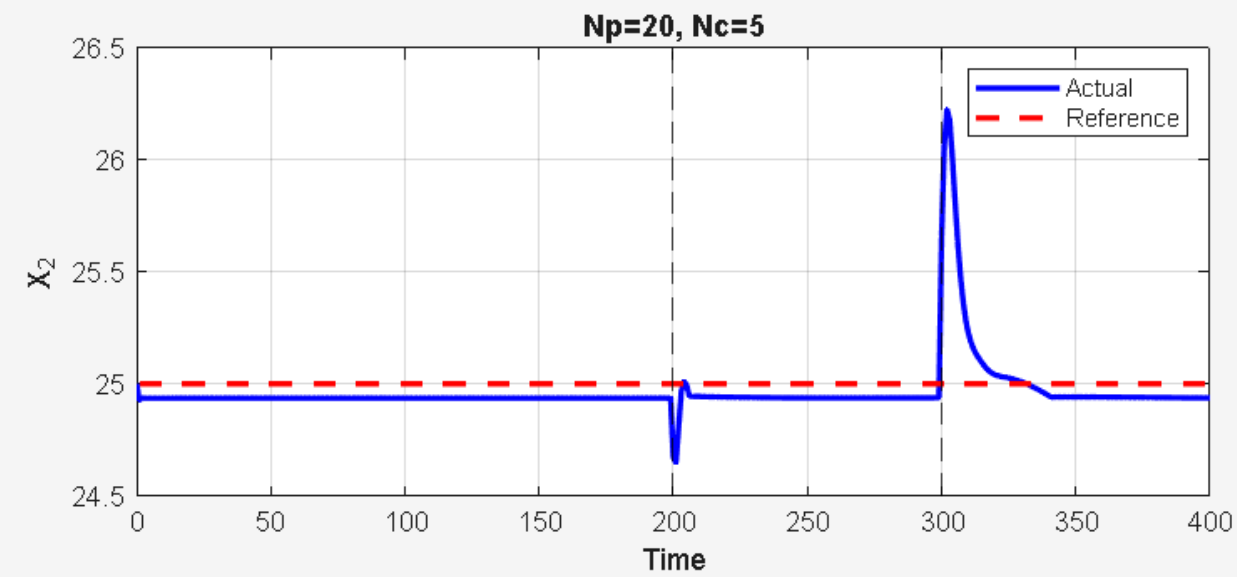


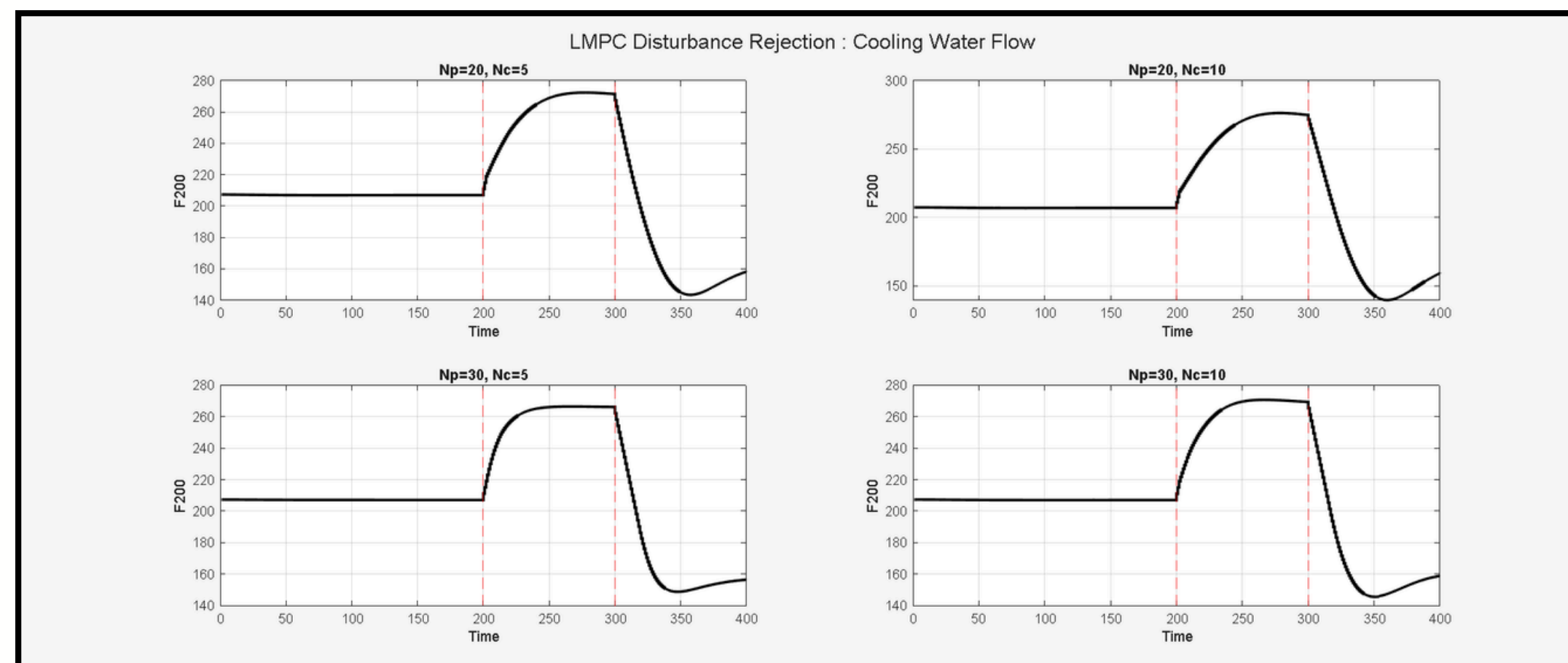
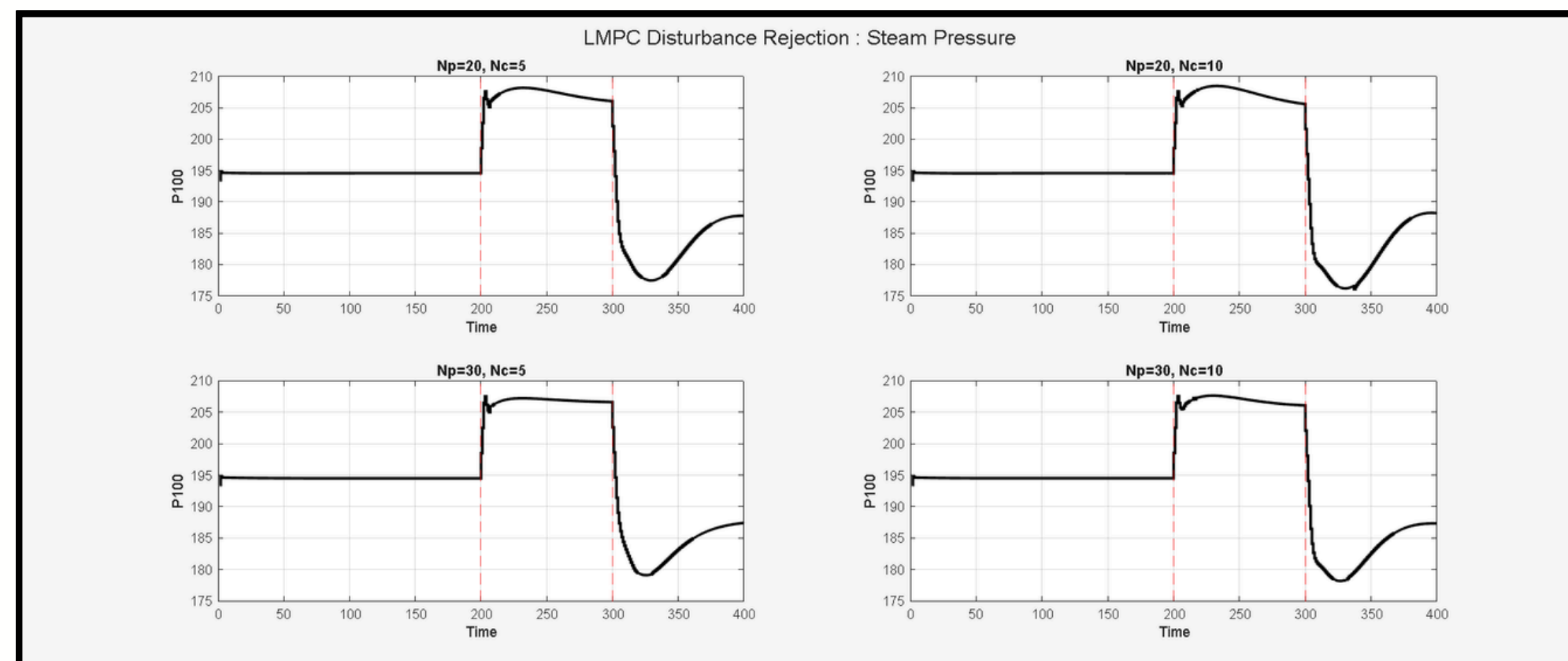
NMPC Setpoint Tracking : Cooling Water Flow



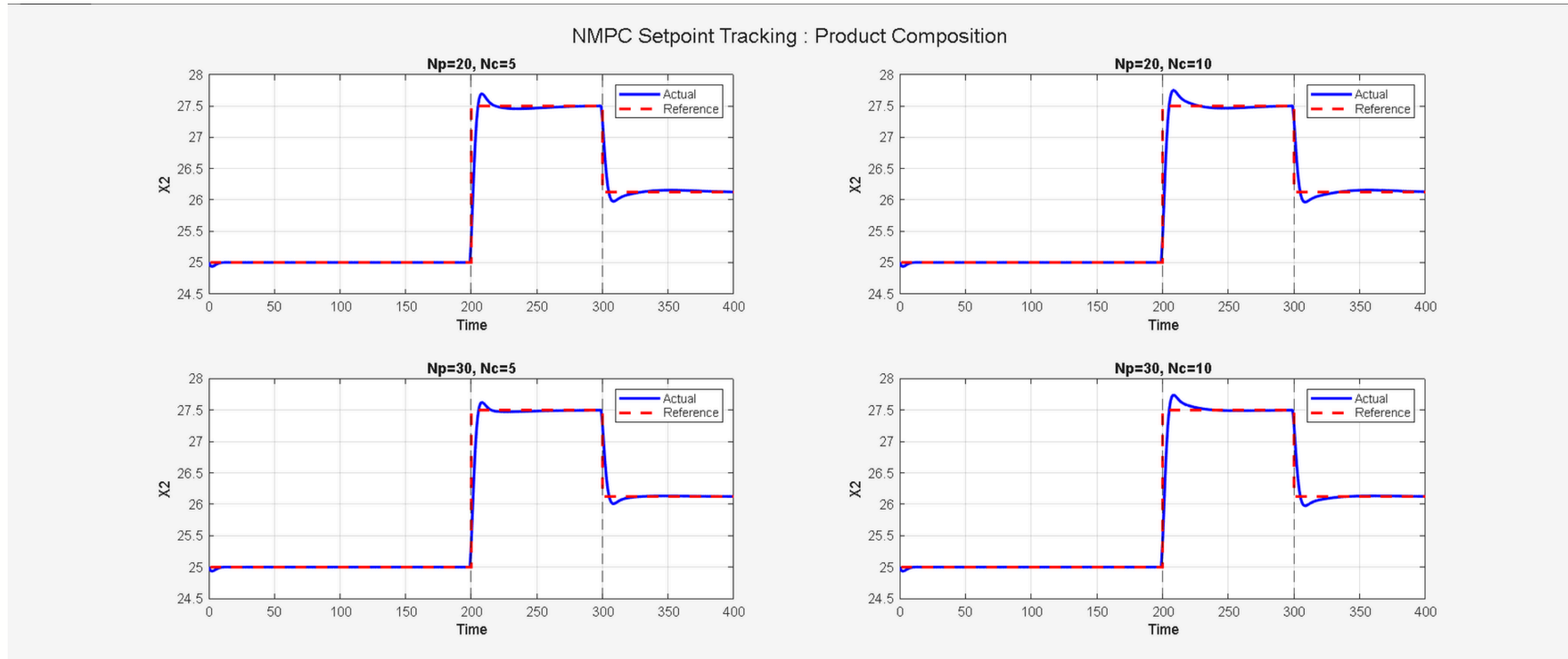
LMPC Disturbance Rejection – Varying Horizons

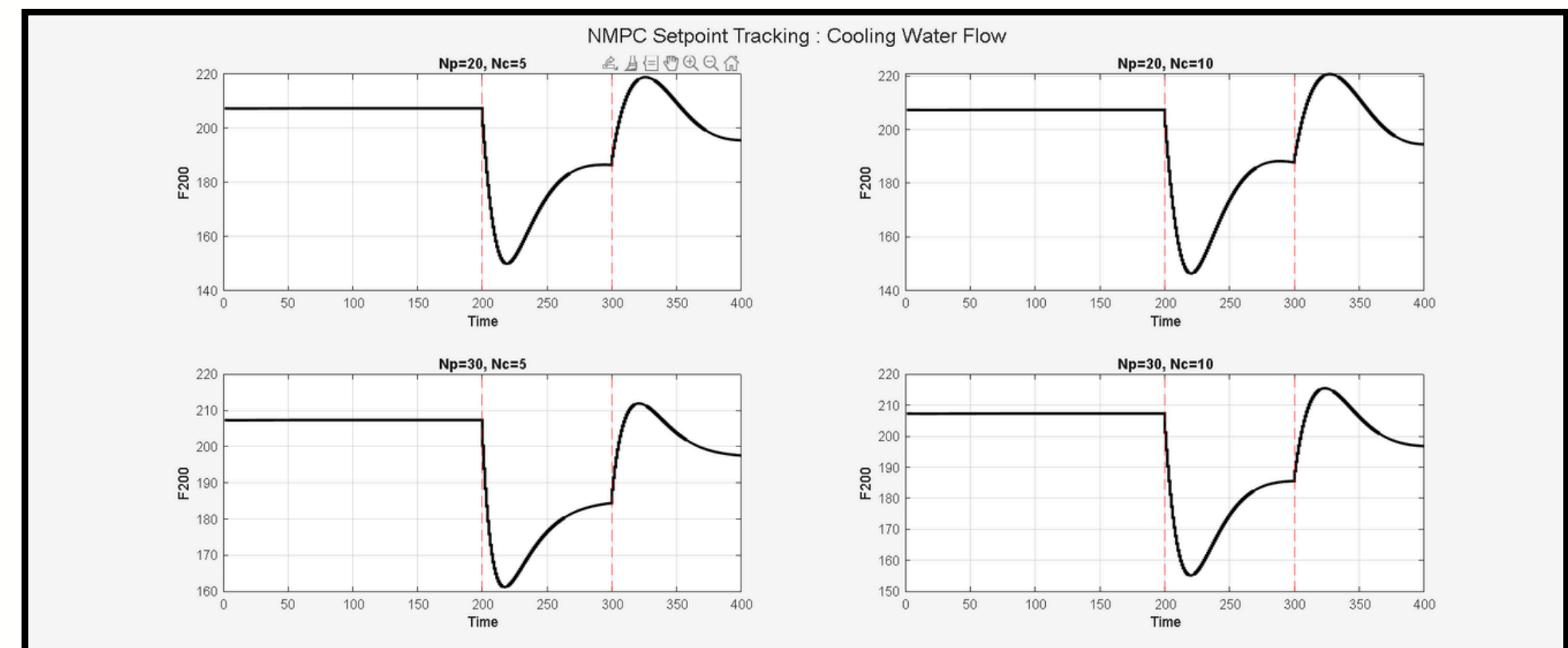
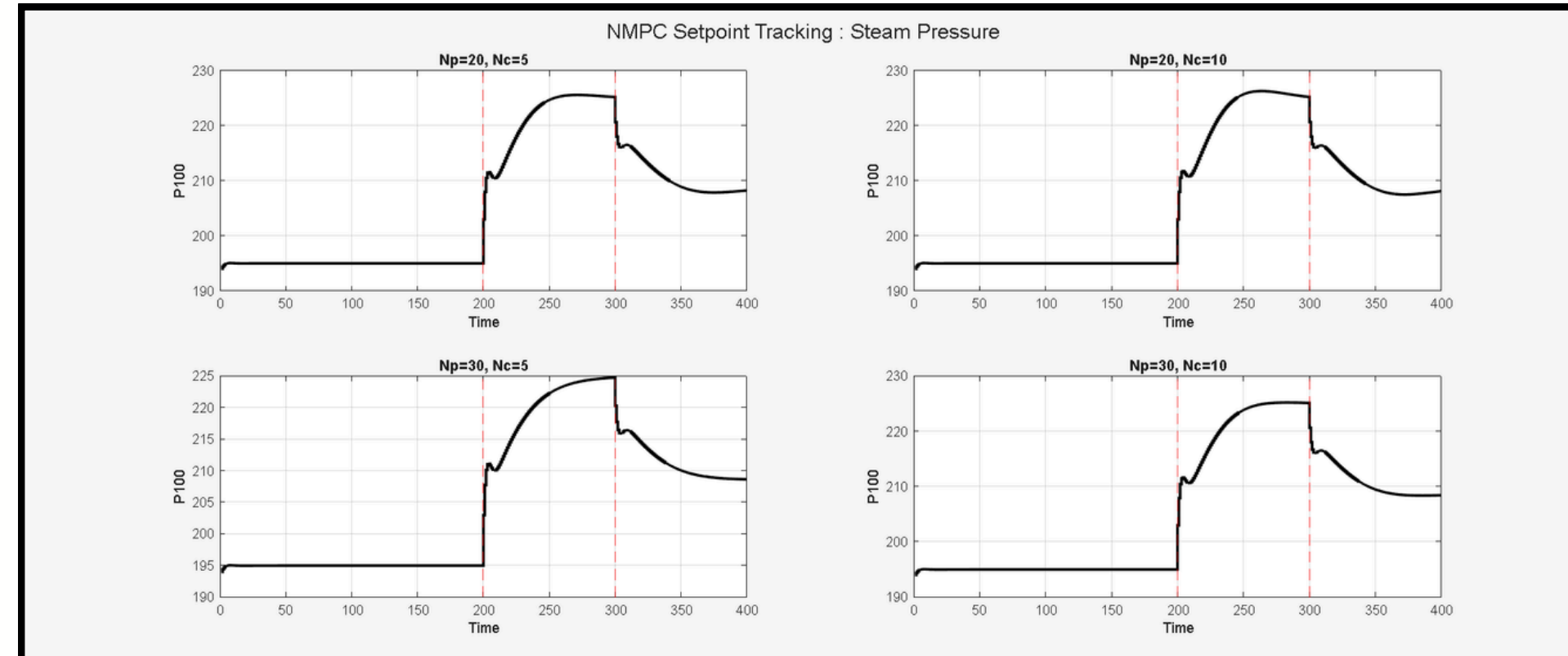
LMPC Disturbance Rejection : Product Composition





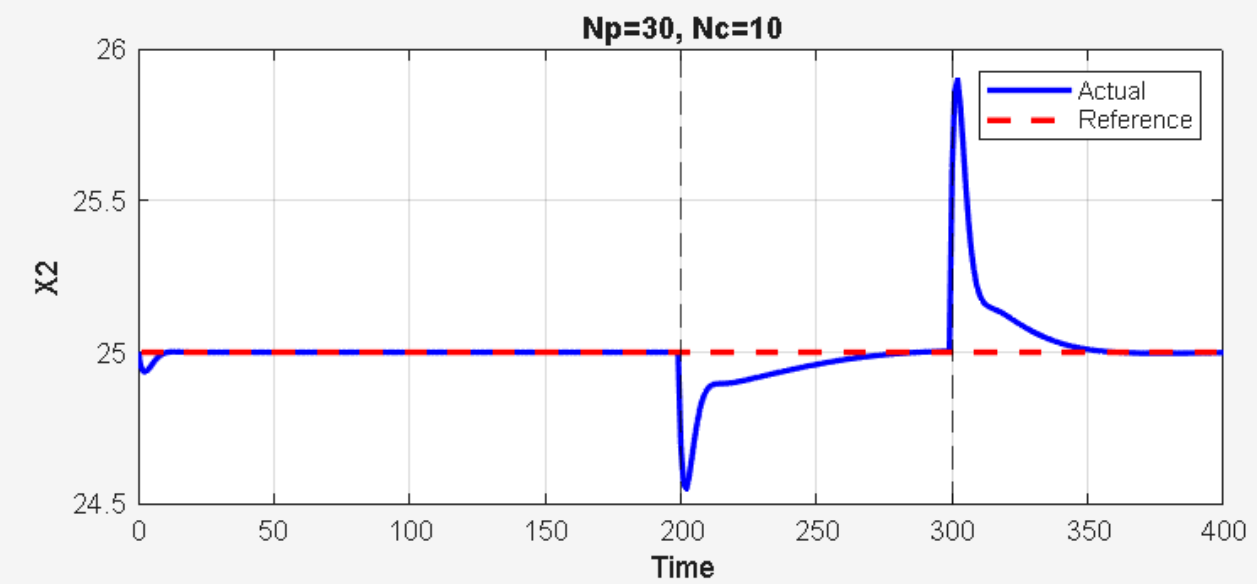
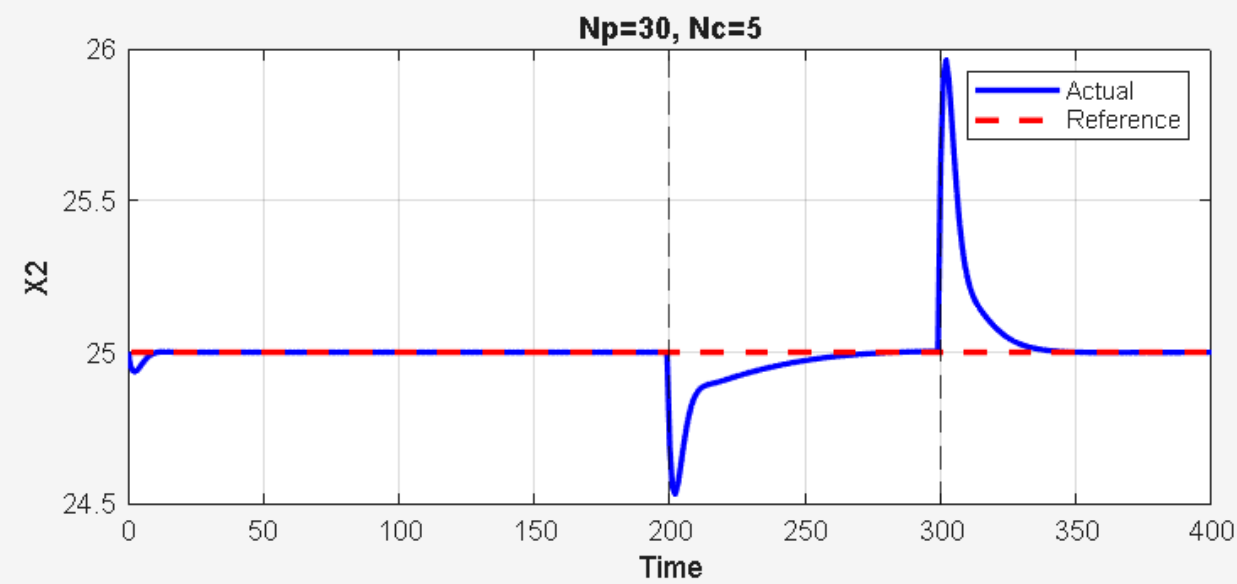
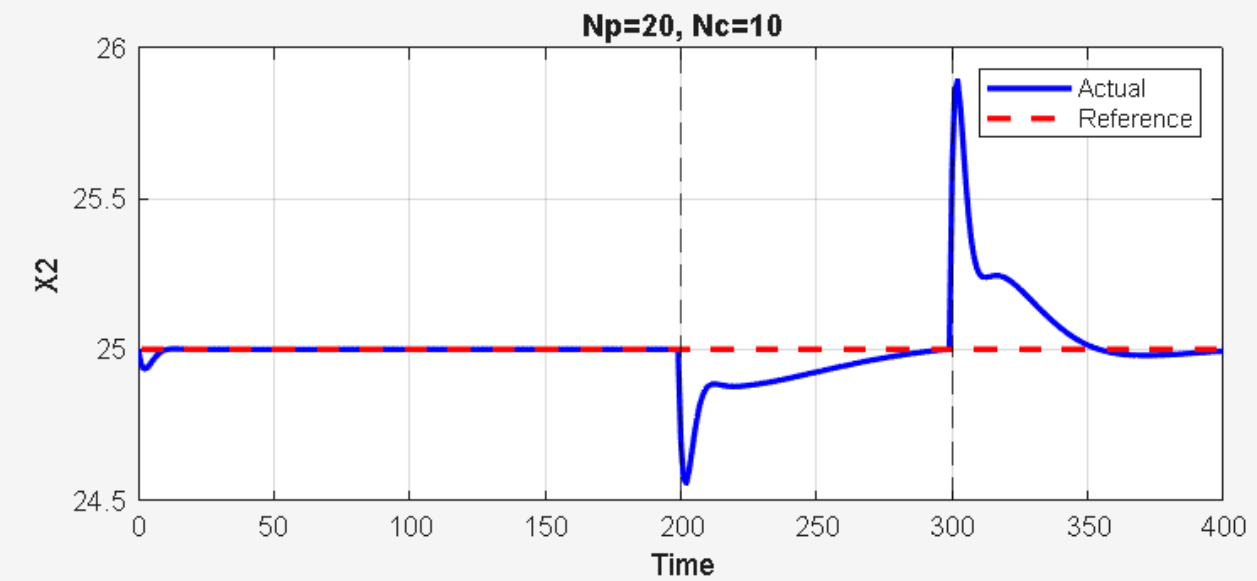
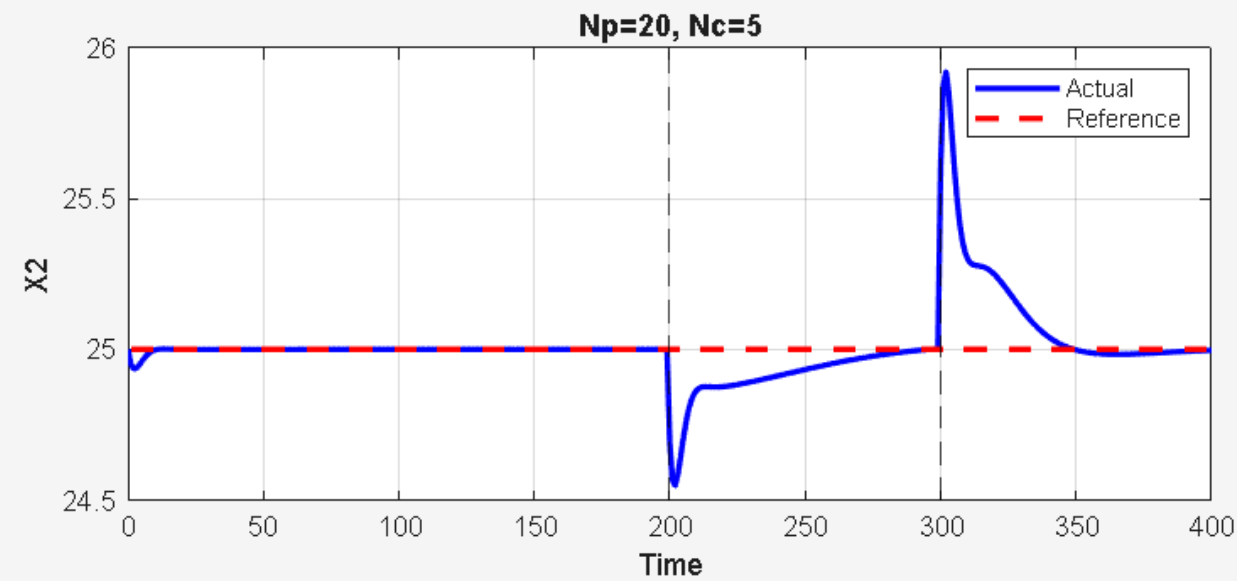
NMPC Set Point Tracking – Varying Horizons



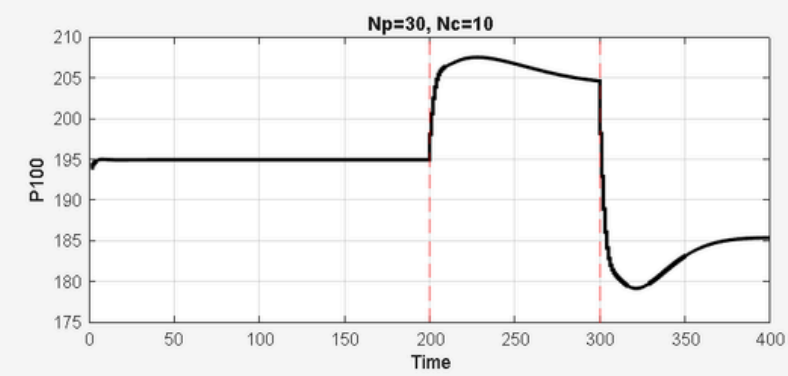
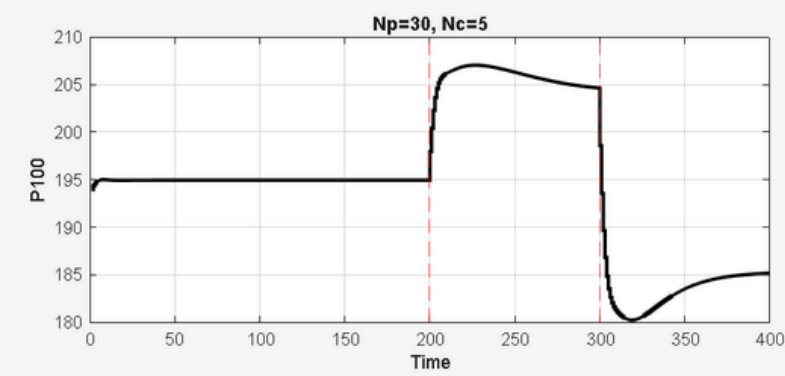
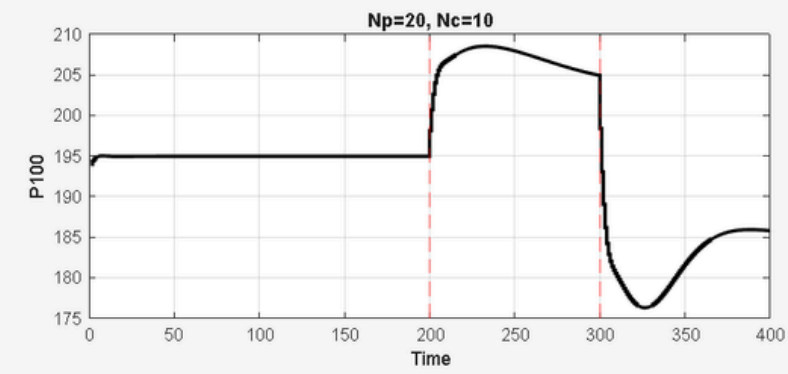
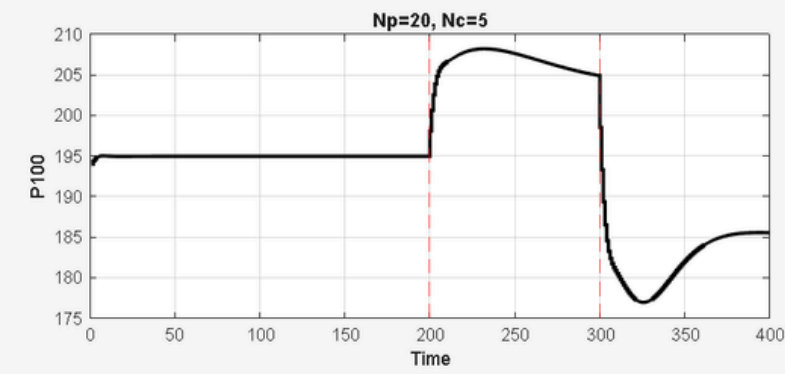


NMPC Disturbance Rejection - Varying Horizons

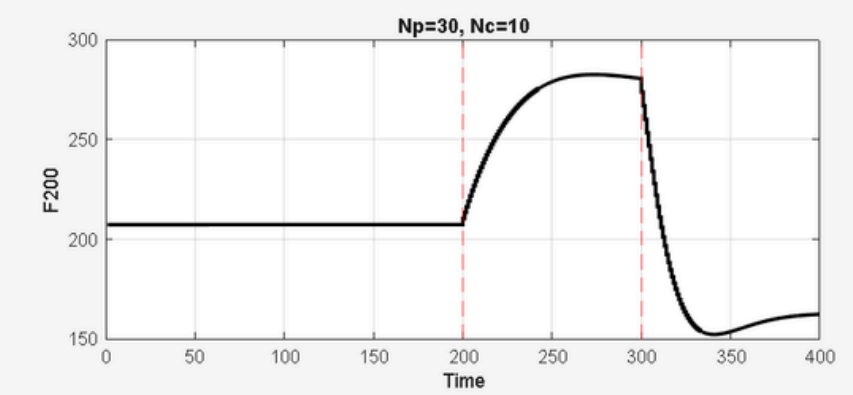
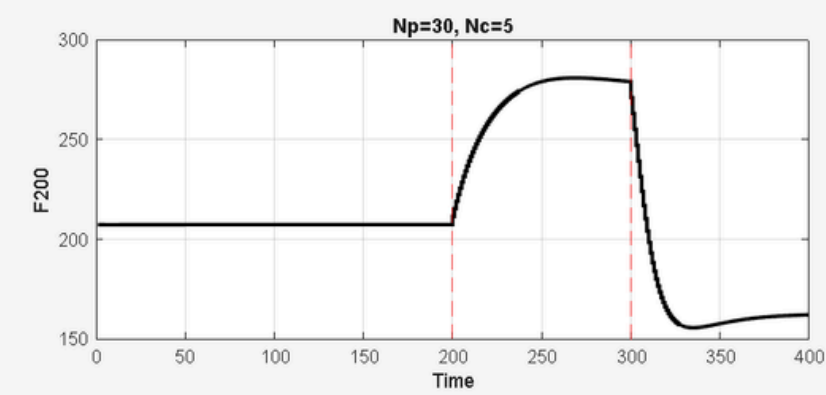
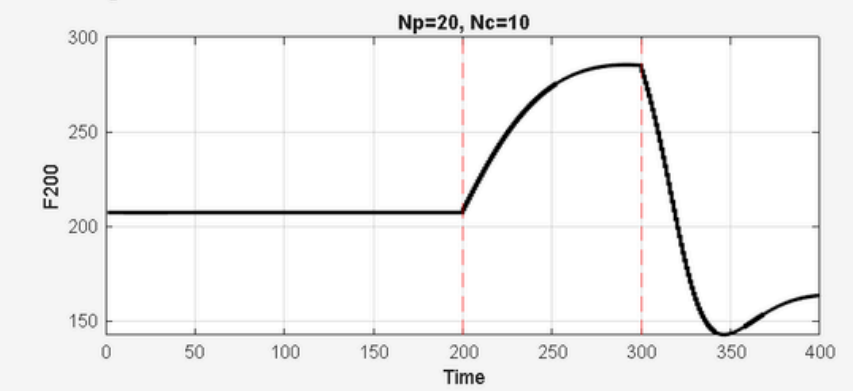
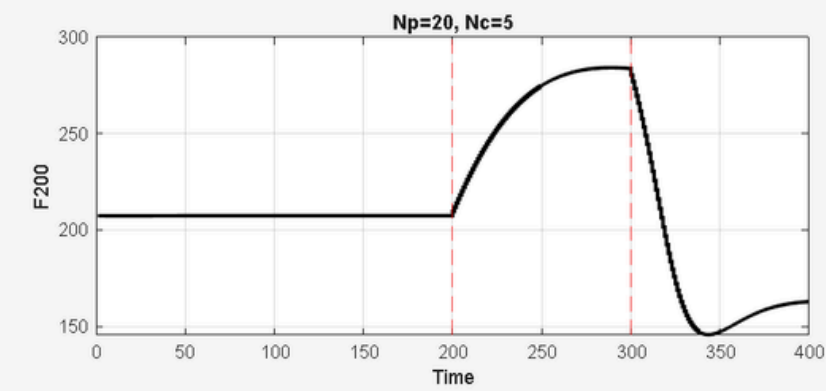
NMPC Disturbance Rejection : Product Composition



NMPC Disturbance Rejection : Steam Pressure



NMPC Disturbance Rejection : Cooling Water Flow



MPC Formulation – Varying Weights

The Model Predictive Controller (LMPC) was implemented for both:

- Disturbance Rejection
- Setpoint Tracking

The controller performance was analyzed by varying the weighting matrices in the MPC objective function:

- State Tracking Weight Matrix (Q)
- Move Suppression Weight Matrix (R)

Weight Matrices Considered

1. State Tracking Weight Matrices

$$Q_1 = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

$$Q_2 = \begin{bmatrix} 20 & 0 \\ 0 & 3 \end{bmatrix}$$

$$Q_3 = \begin{bmatrix} 60 & 0 \\ 0 & 8 \end{bmatrix}$$

2. Move Suppression Weight Matrices

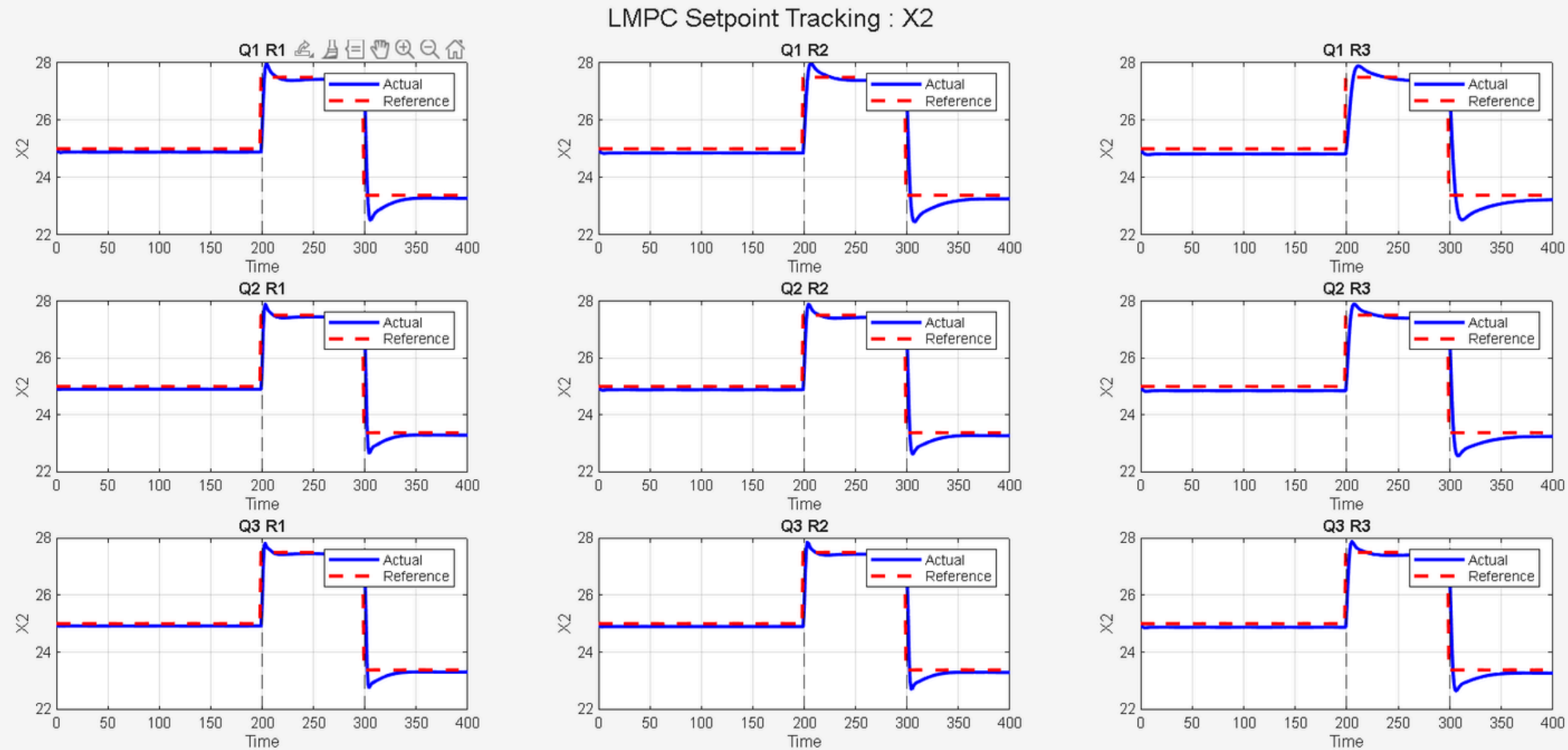
$$R_1 = \begin{bmatrix} 0.05 & 0 \\ 0 & 0.05 \end{bmatrix}$$

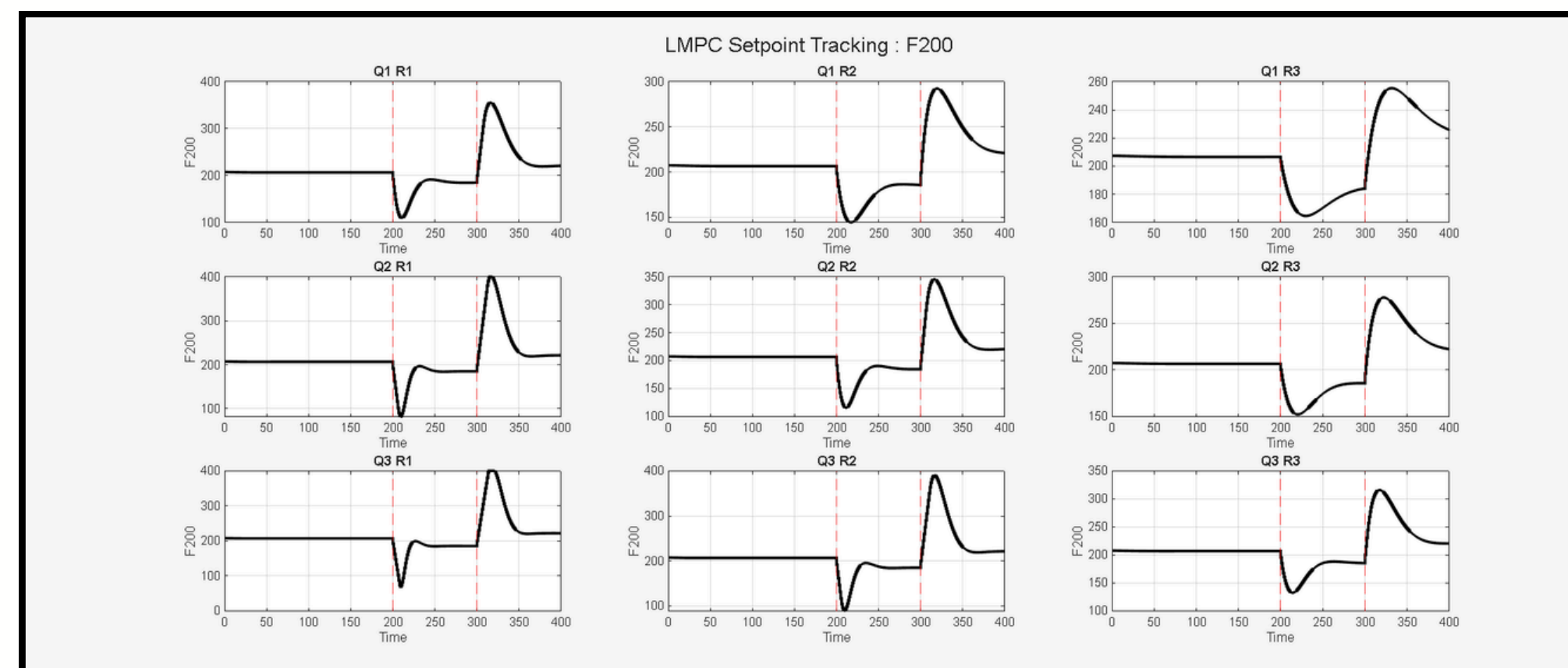
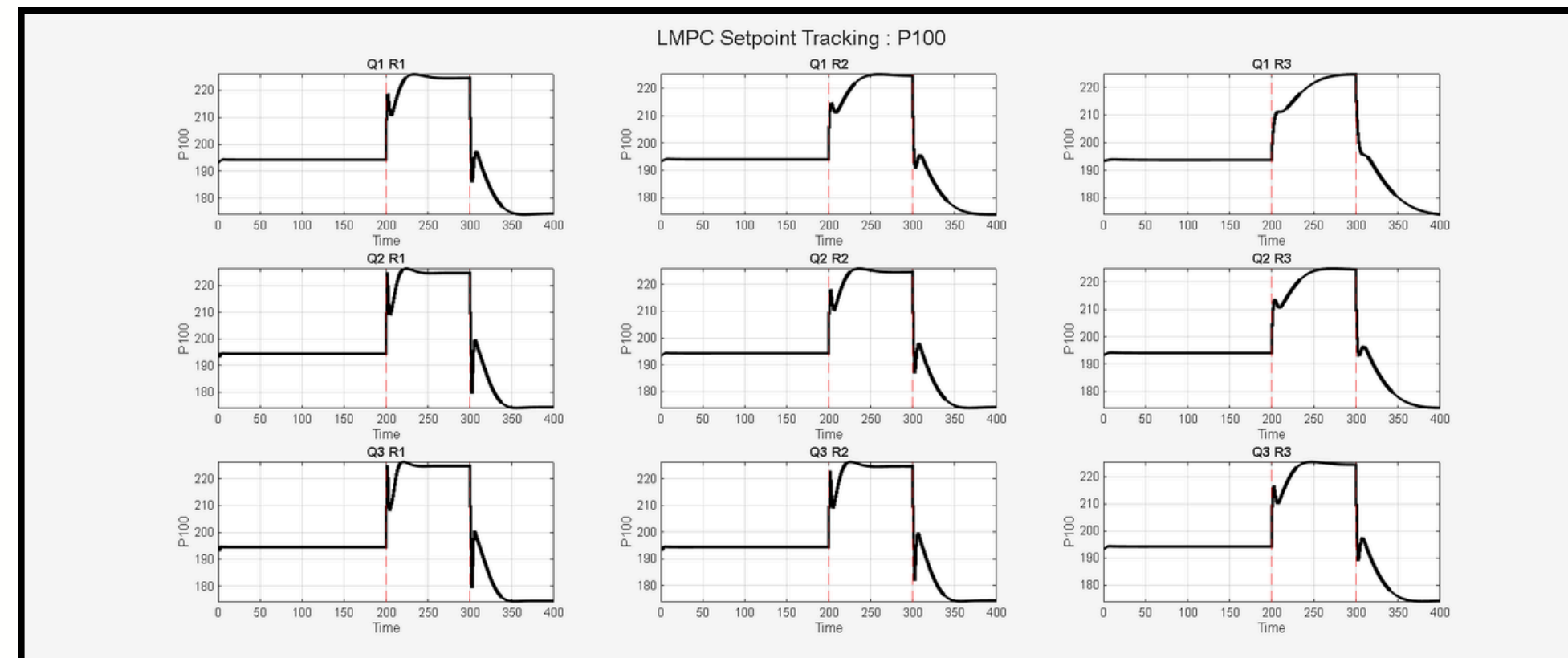
$$R_2 = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \end{bmatrix}$$

$$R_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

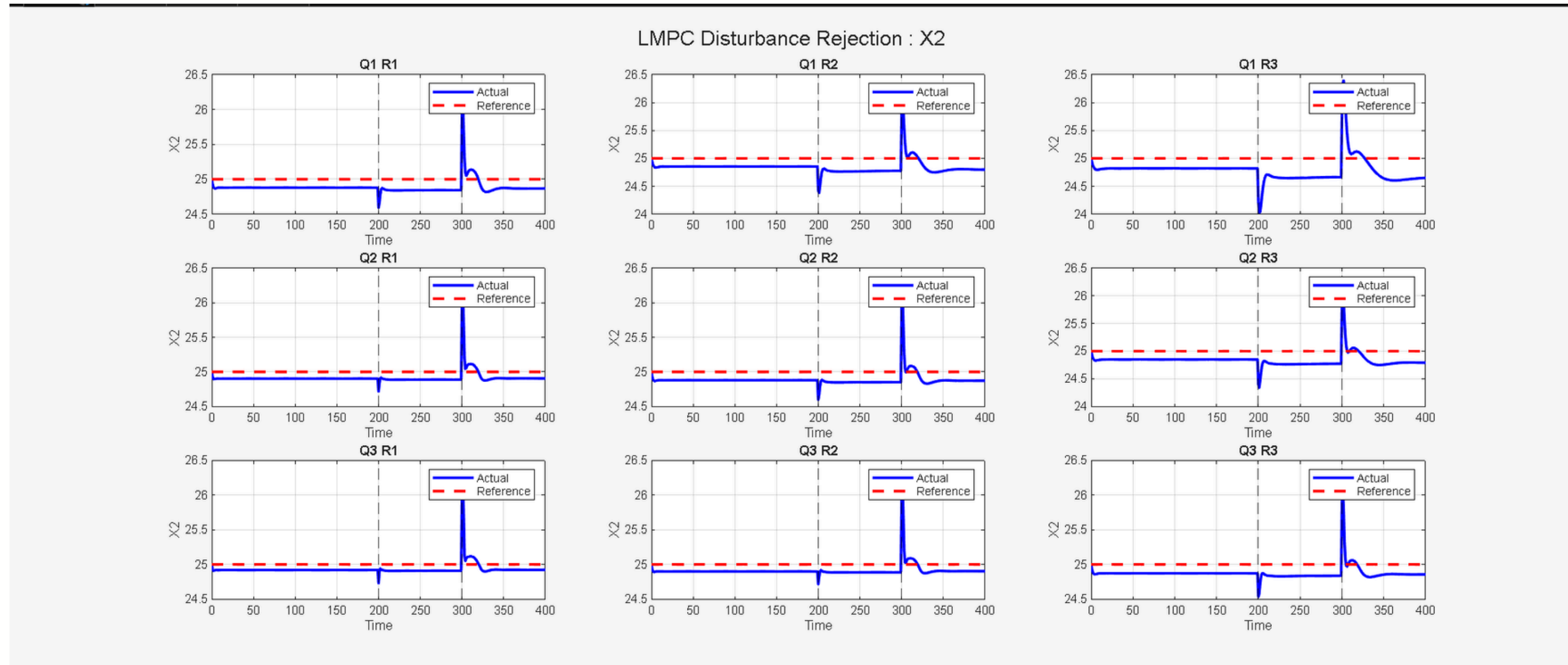
Optimization solved using MATLAB fmincon with Sequential Quadratic Programming (SQP).

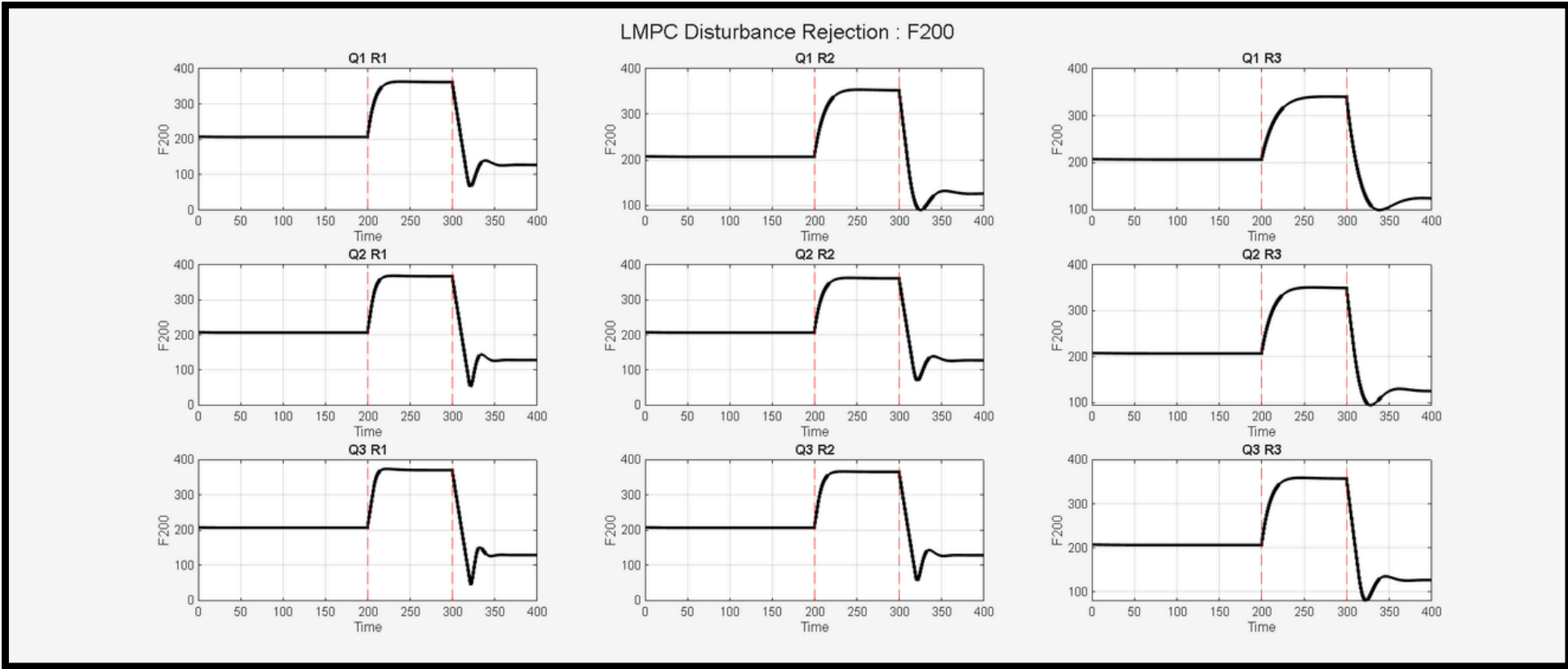
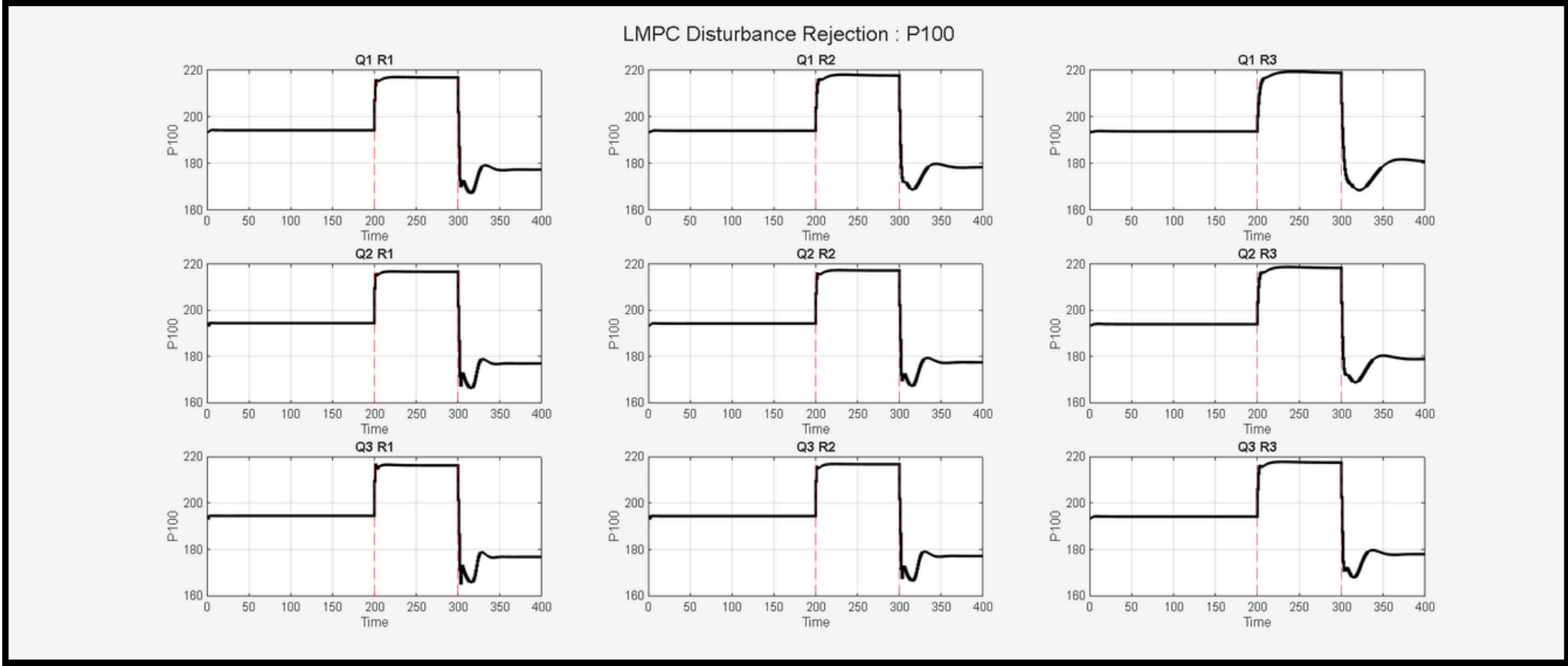
LMPC Set Point Tracking - Varying Weights



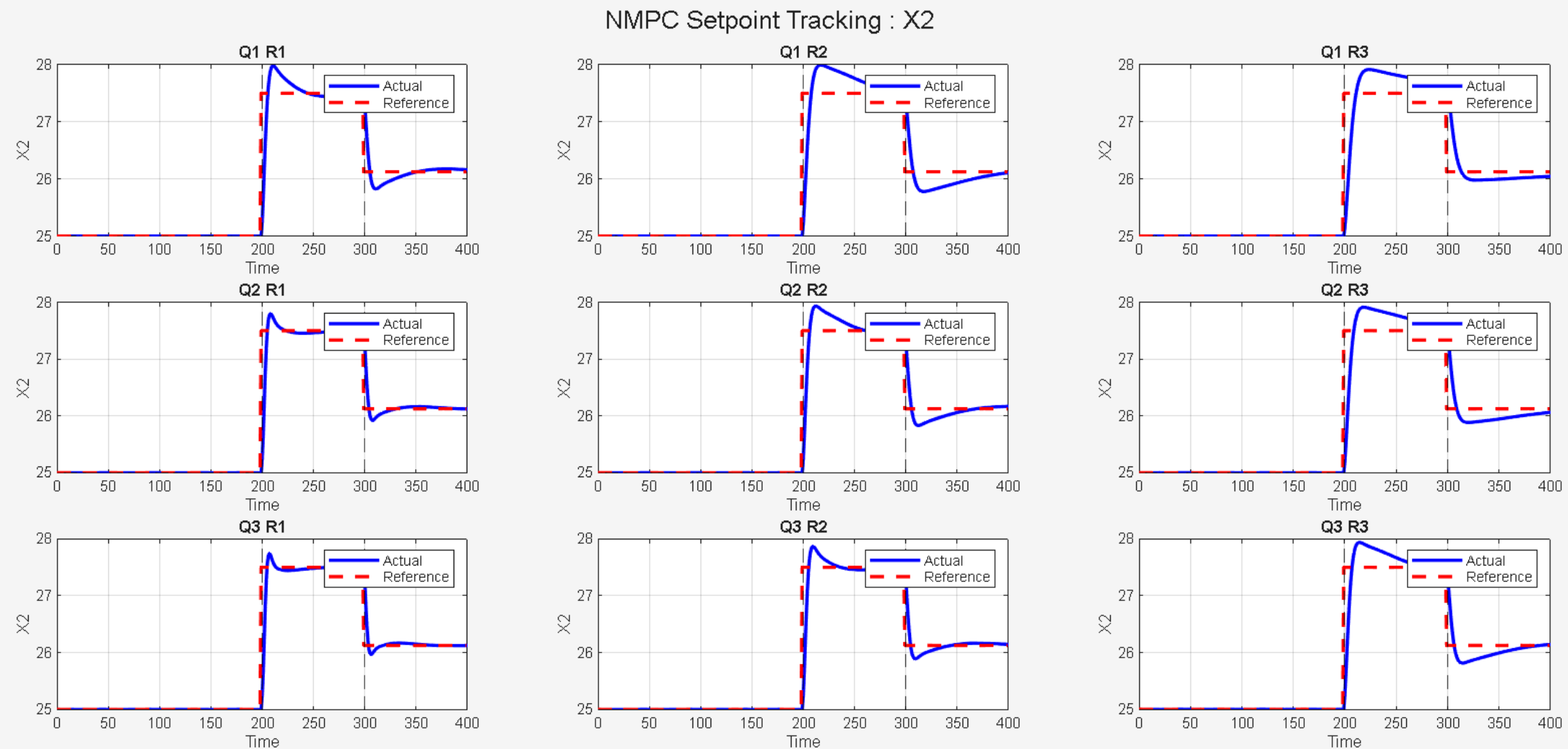


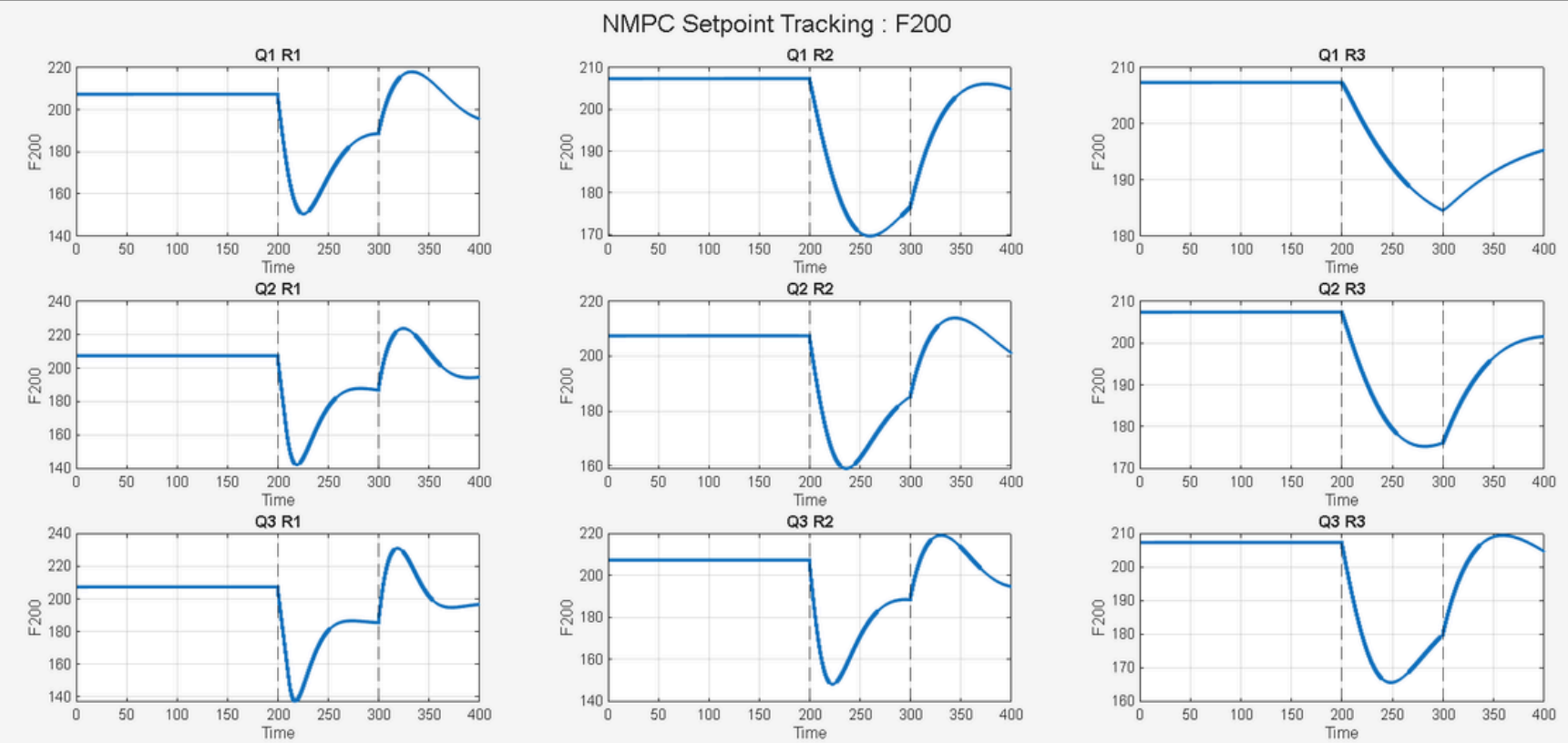
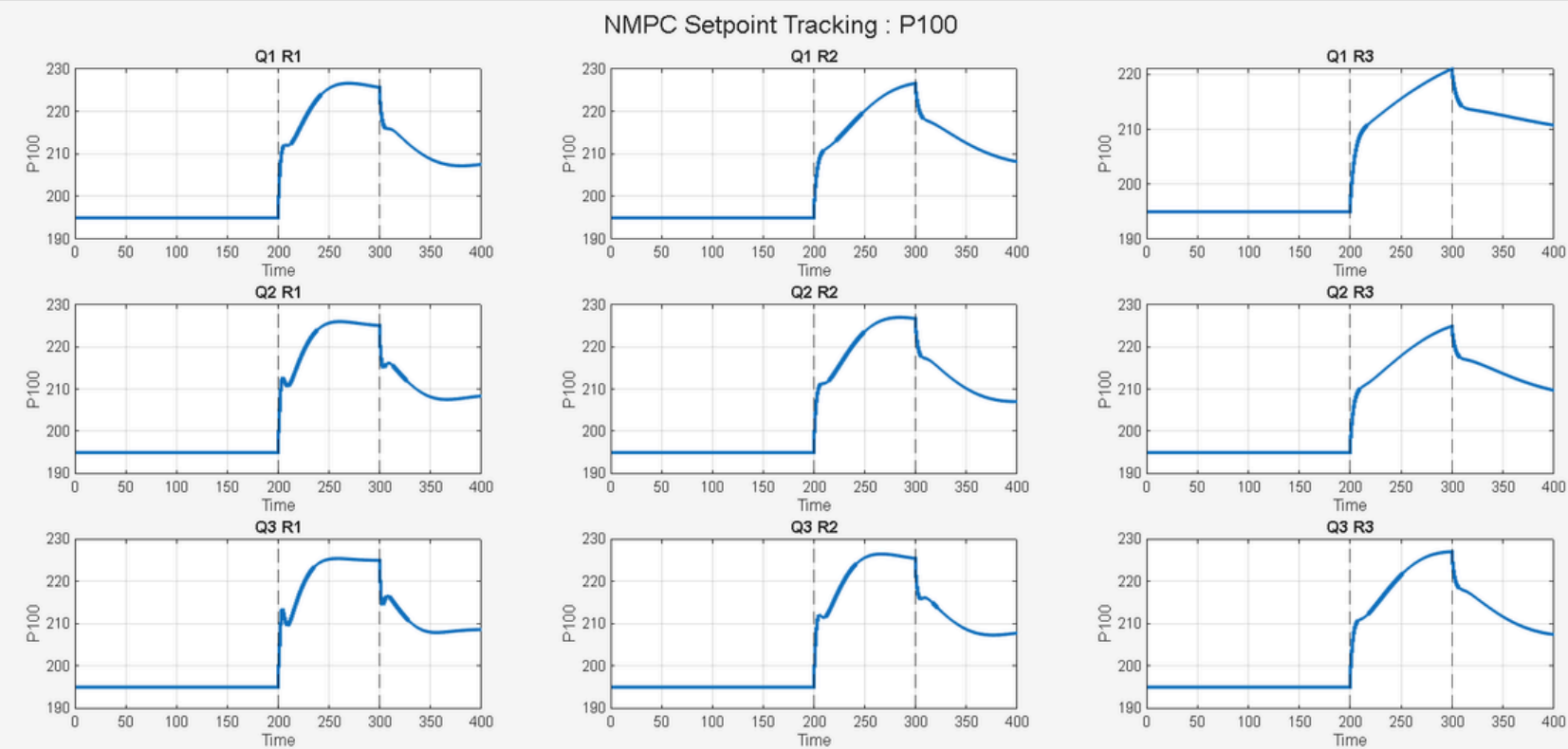
LMPC Disturbance Rejection – Varying Weights



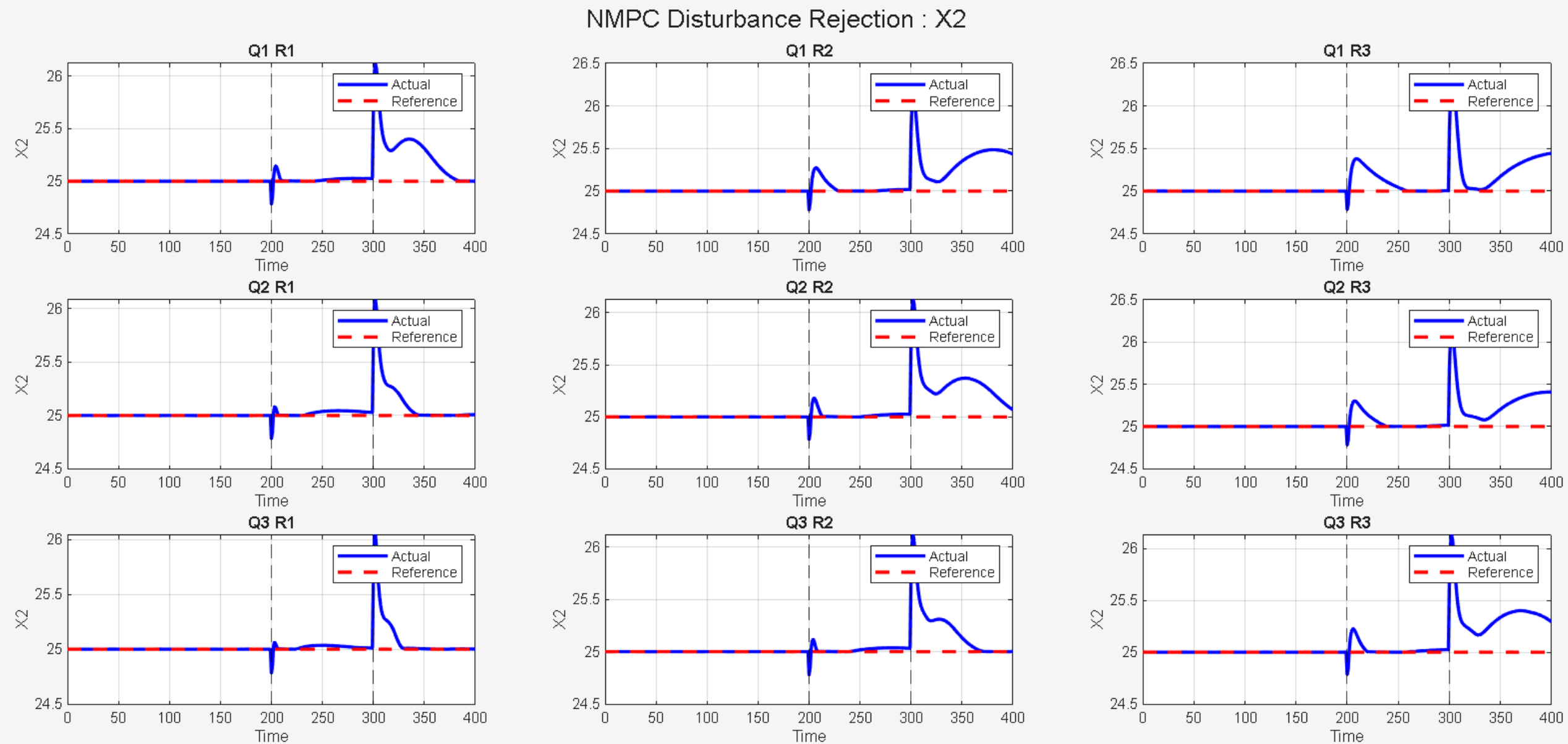


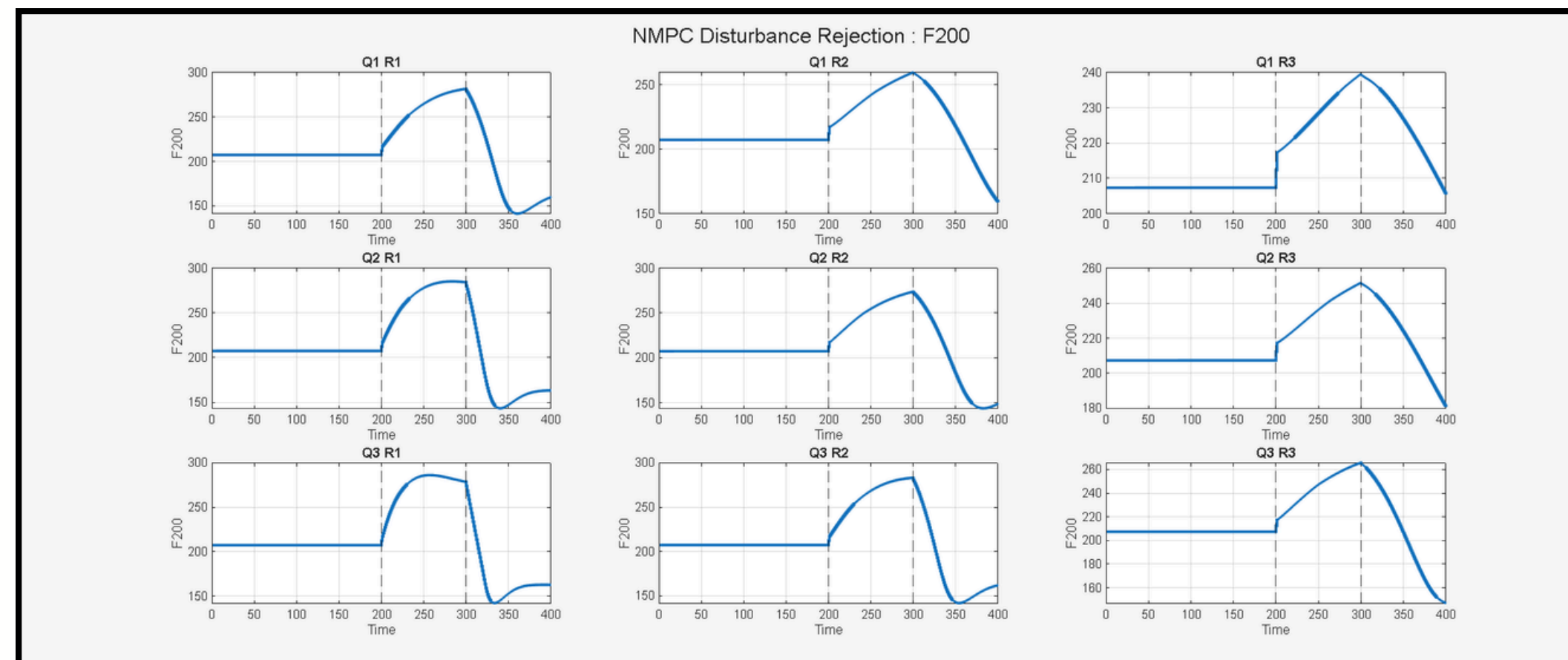
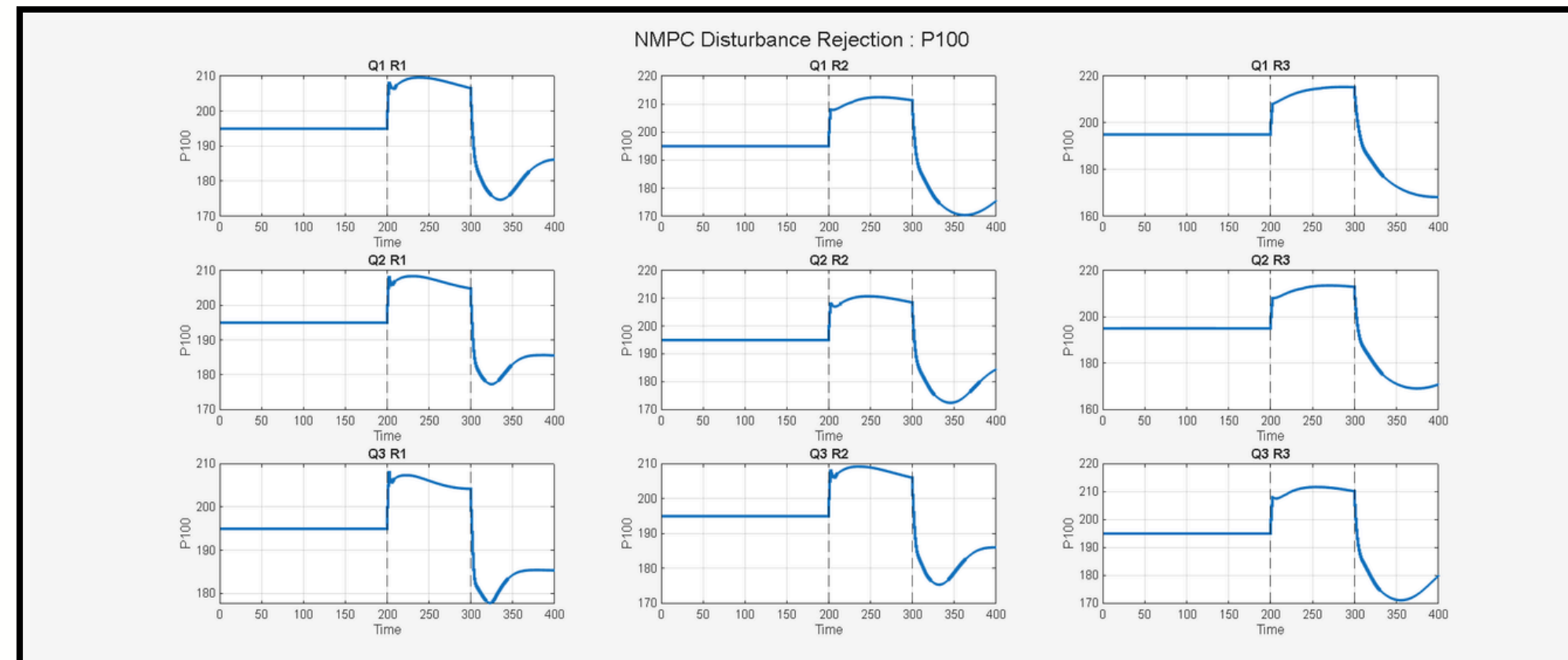
NMPC Set Point Tracking - Varying Weights



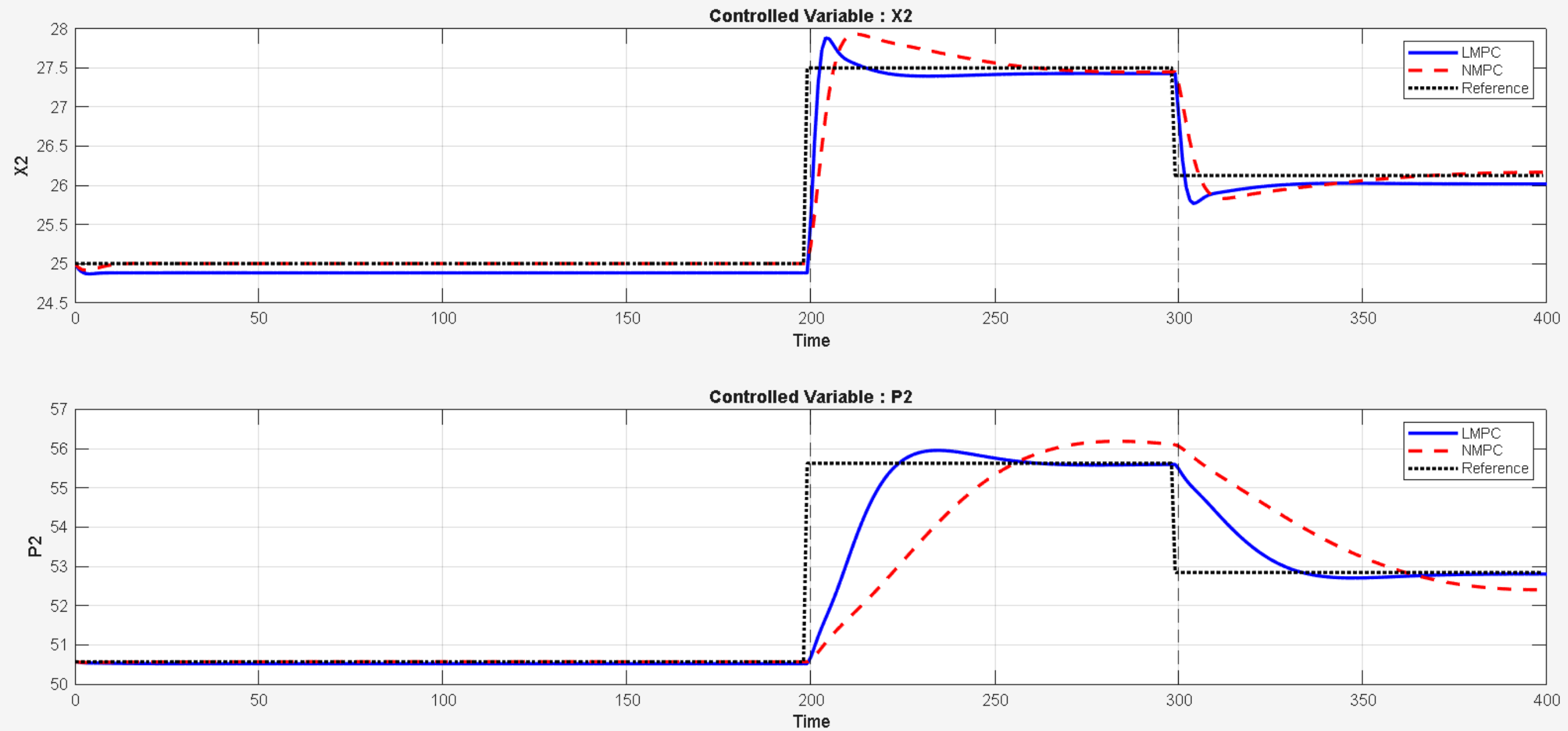


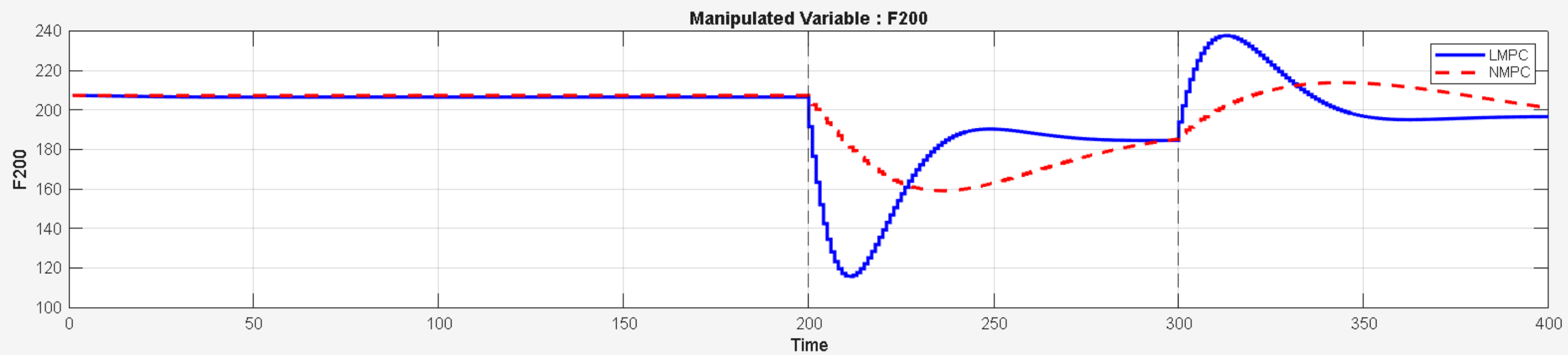
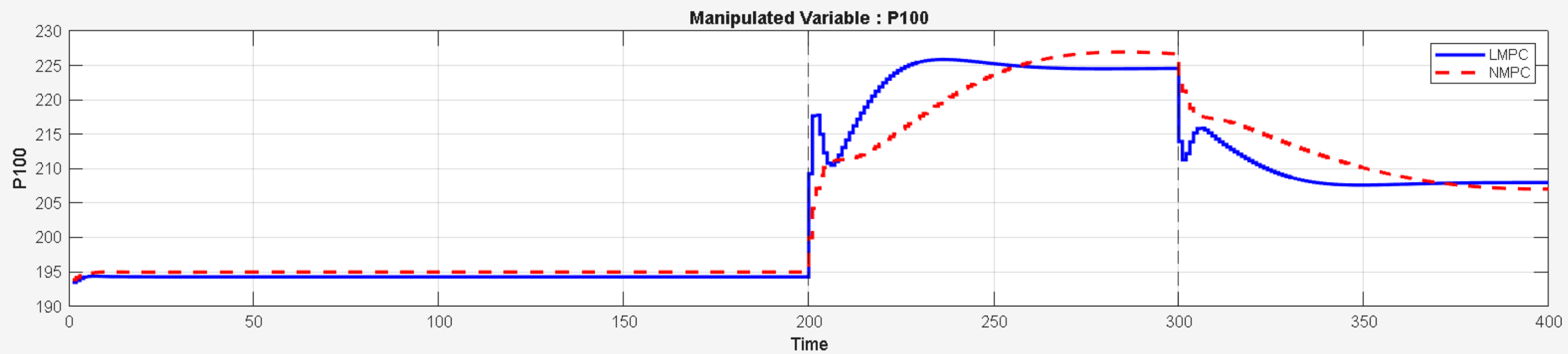
NMPC Disturbance Rejection – Varying Weights



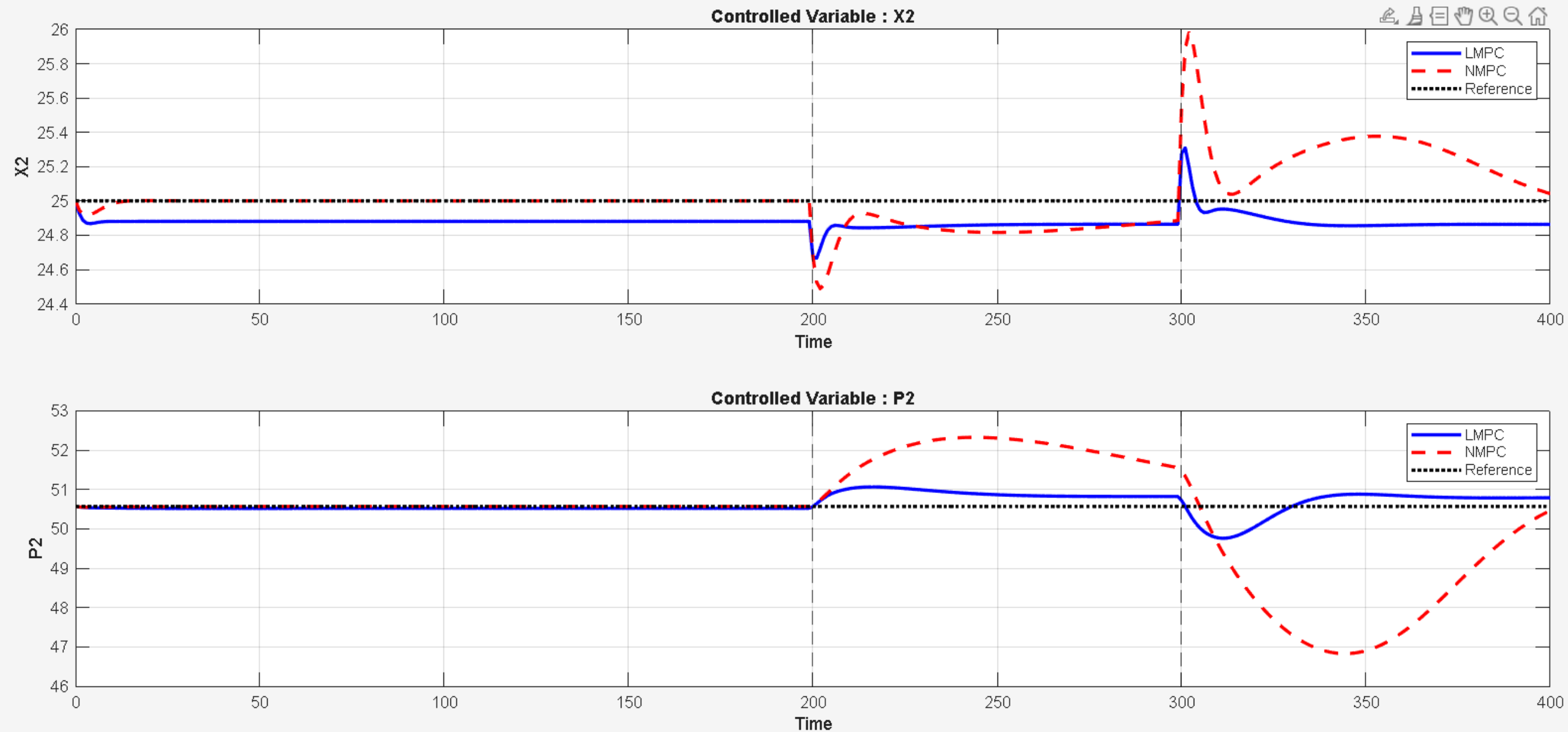


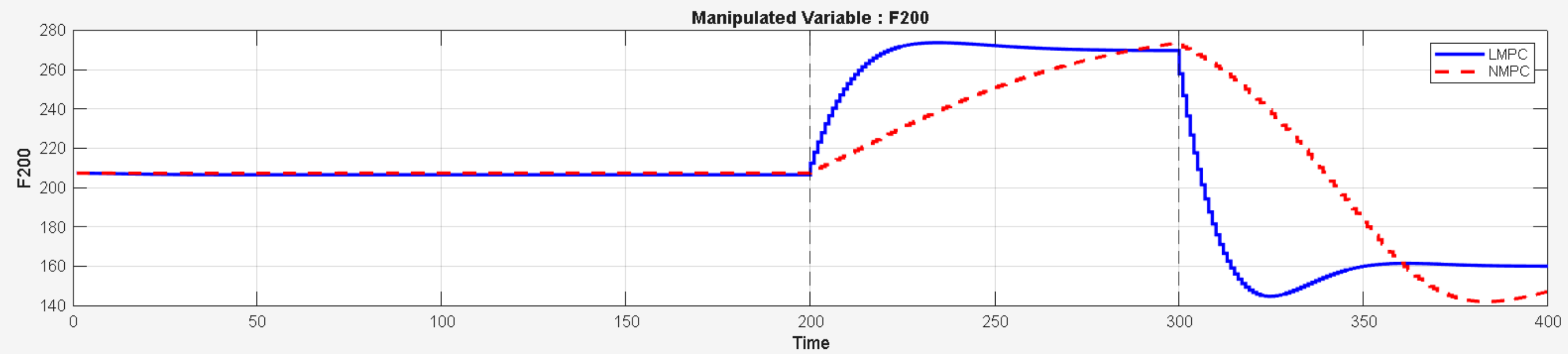
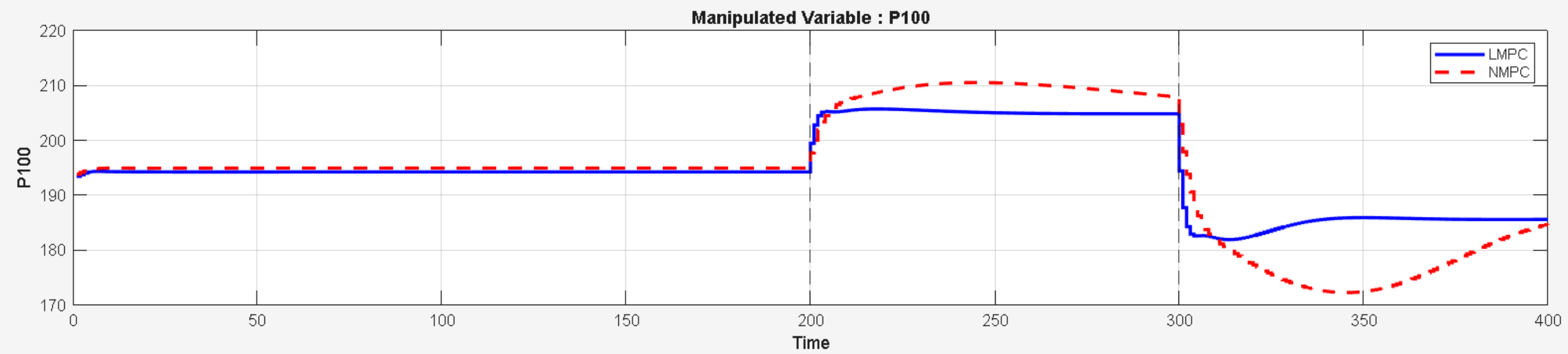
Comparison LMPC vs NMPC – Set Point Track





Comparison LMPC vs NMPC – Disturbance Rej.





Conclusion

- Both LMPC and NMPC successfully maintained the desired product concentration X_2 and operating pressure P_2 within the specified constraints.
- NMPC outperformed LMPC in disturbance rejection with faster settling and minimal oscillations, while LMPC showed larger deviations after disturbances.
- NMPC achieved better setpoint tracking accuracy due to its nonlinear model, but at the cost of higher computational effort and slower response.
- Higher Q (state weight) improved tracking but caused aggressive control action; higher R gave smoother inputs with slight steady-state offset.



Thank You